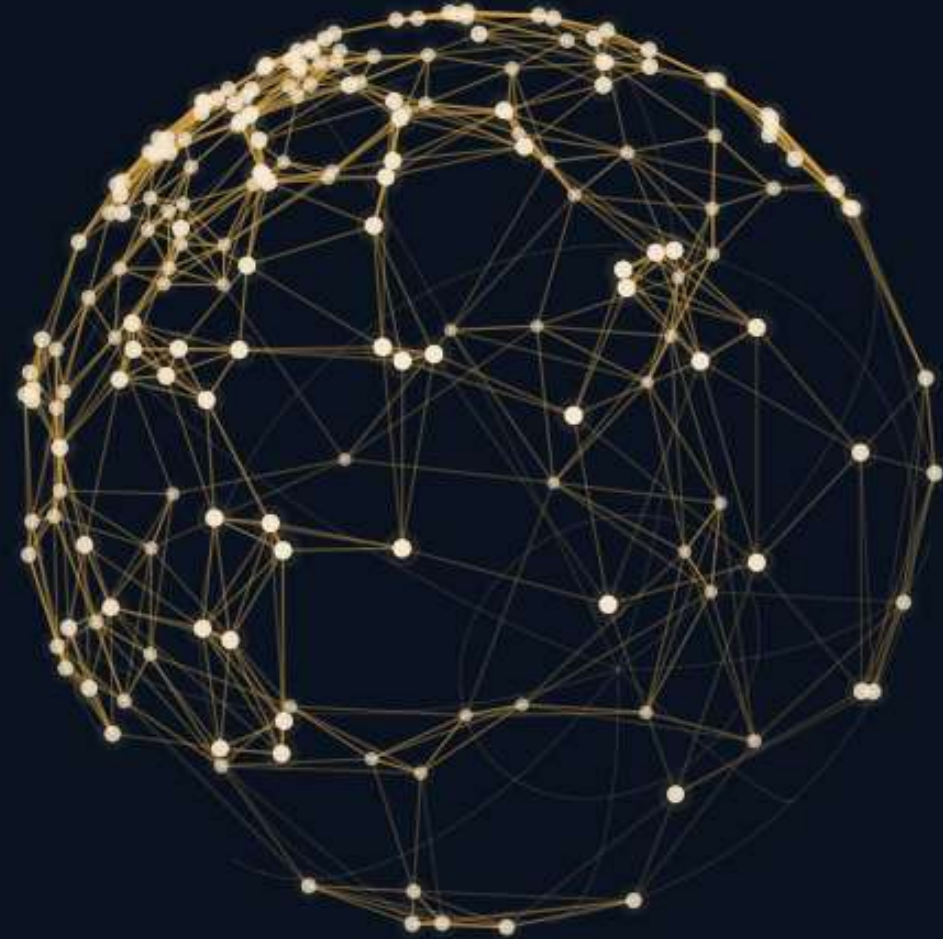


Closed Universes, Wormholes, and Quantum Magic

Brian Swingle (Brandeis)
UChicago Seminar, April 27, 2026

With [Valerie Bettaque](#) (2602.12339)
and [Alejandro Vilar Lopez](#) and [Martin Sasieta](#) (2512.00149)





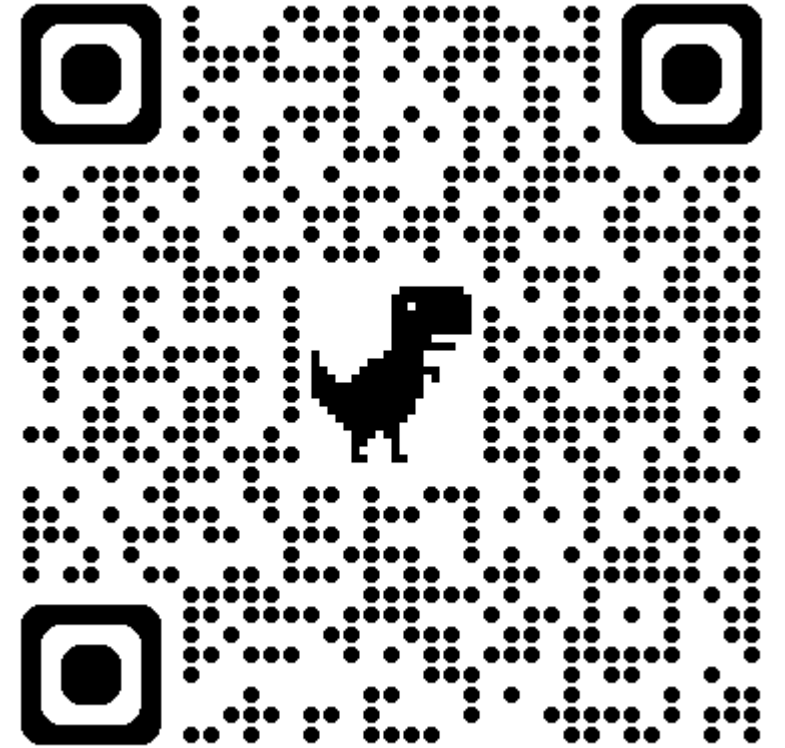
Valerie Bettaque



Alejandro Vilar Lopez



Martin Sasieta



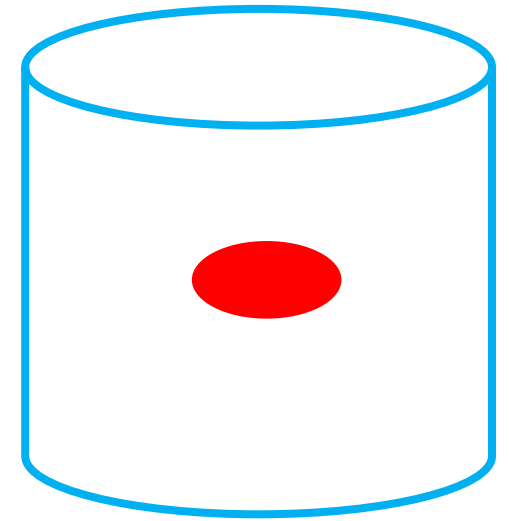
physicsmonkey.github.io

Introduction

AdS/CFT and “it from qubit”

For a “universe in a bottle” (Anti de Sitter space), the AdS/CFT correspondence gives a framework for quantum gravity: [Maldacena, ...]

- Wonderful theoretical lab to study black holes, strong coupling dynamics, hydrodynamics, ...
- Gravitational physics is intimately linked with quantum information: entanglement and geometry, complexity and black holes, ...
- By simulating the boundary system in a lab, we can hope to experimentally study holography!



Stuff in AdS $\leftrightarrow$
States in dual CFT

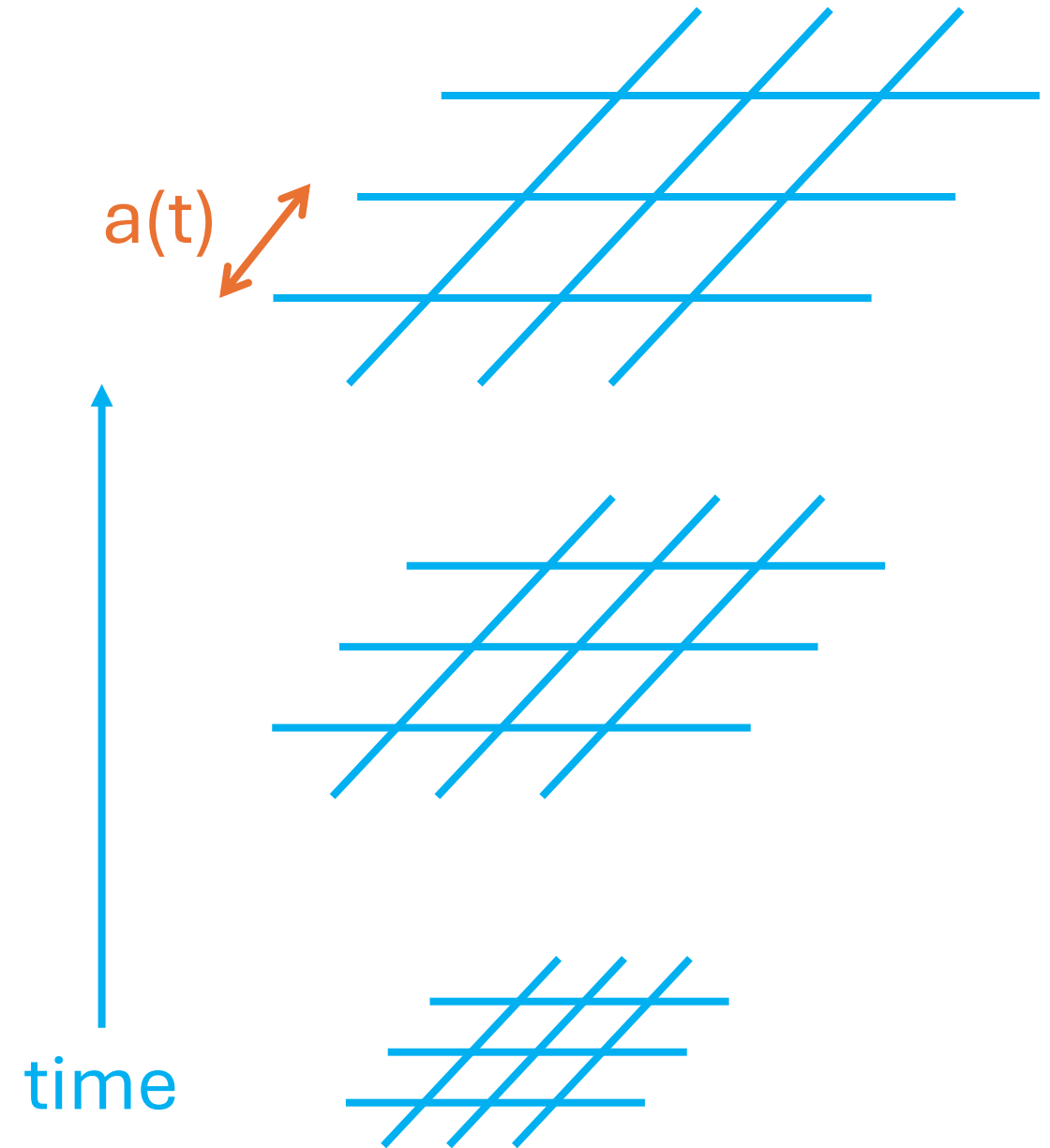
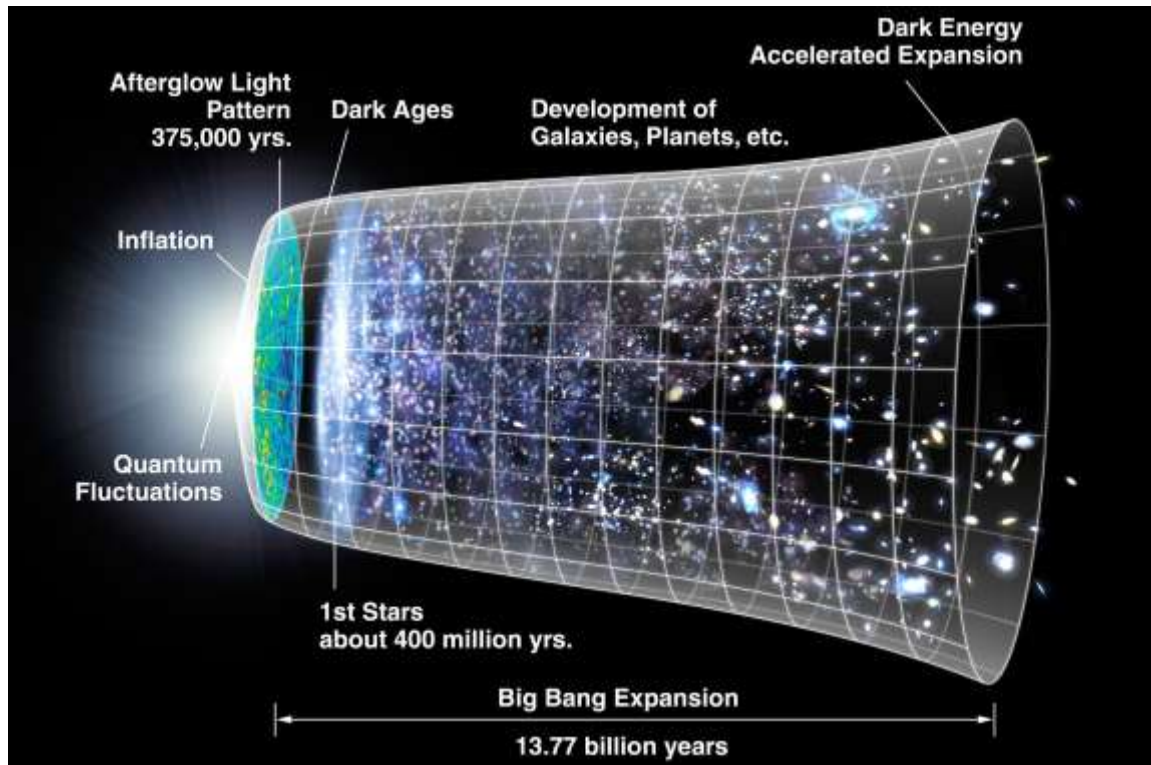
GR and QM together!

- This form of holography is not directly relevant to quantum gravity in our world, but it serves as a powerful proof of principle and theoretical lab
- It also connects to myriad phenomena in quantum matter and quantum info
- Today I want to tell you about progress towards a quantum gravitational description of our universe



Our universe is:

- Full of stuff, homogenous, ...
- Expanding: $a(t)$
- Λ CDM: positive dark energy
- **No AdS boundaries**



What is the quantum gravitational description of this kind of cosmological physics?

One starting point: gravitational EFTs with negative cosmological constant have such cosmological solutions, so can we understand their meaning in the dual CFT?

- Fine print: this is not the only route; the models so far are toy models; at least we will have a big bang!

Cosmology and wormholes

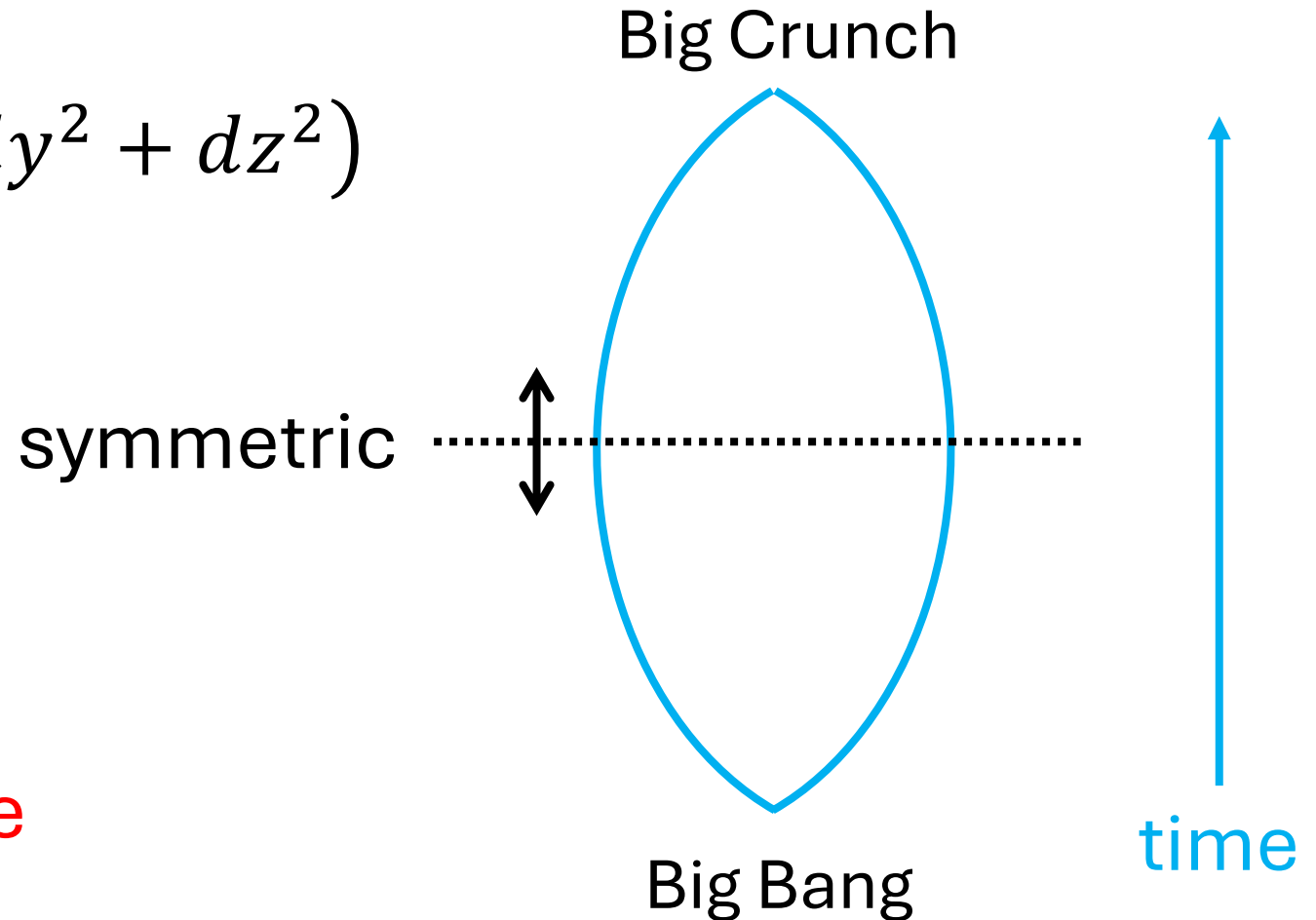
Example

Flat + radiation + $\Lambda < 0$

$$ds^2 = -dt^2 + a(t)^2(dx^2 + dy^2 + dz^2)$$

$$a(t)^2 = \cos t/t_*$$

No AdS boundaries; this is a
flat and homogeneous space



Example

Flat + radiation + $\Lambda < 0$

$$ds^2 = -dt^2 + a(t)^2(dx^2 + dy^2 + dz^2)$$

$$a(t)^2 = \cos t/t_*$$

symmetric



Big Crunch



Add enough bells and whistles (e.g. scalar fields) → you are here??

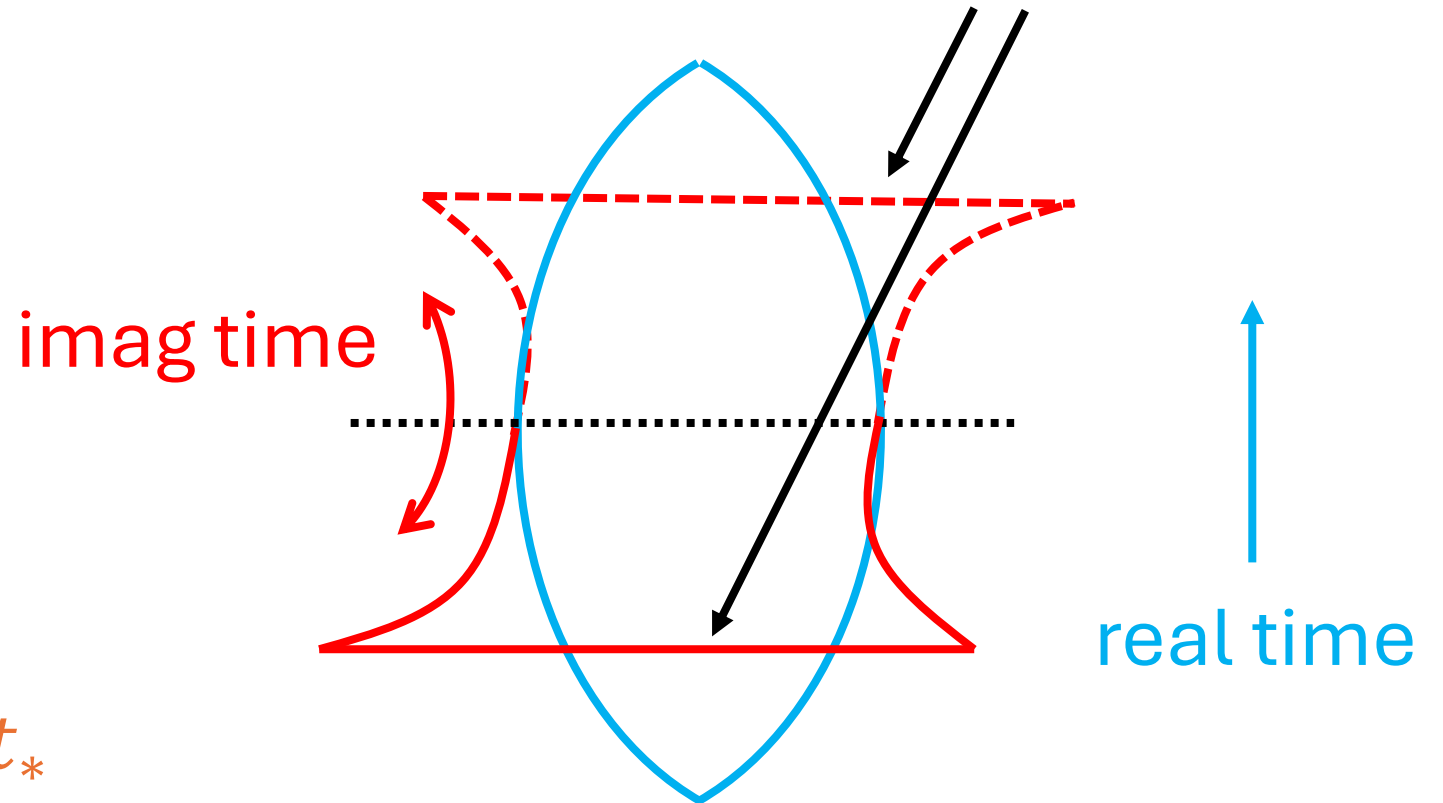
Big Bang

time

[more realistic models: Antonini-Waddell-Simidzija-Van Raamsdonk-S, ...]

Meaning?

Euclidean wormhole connecting
two distinct AdS boundaries

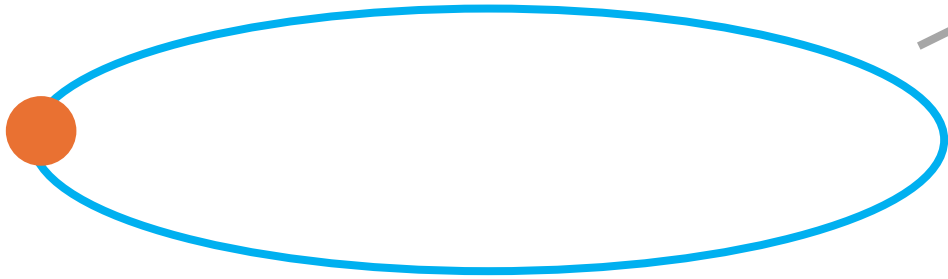


$$a(t)^2 = \cos t/t_*$$

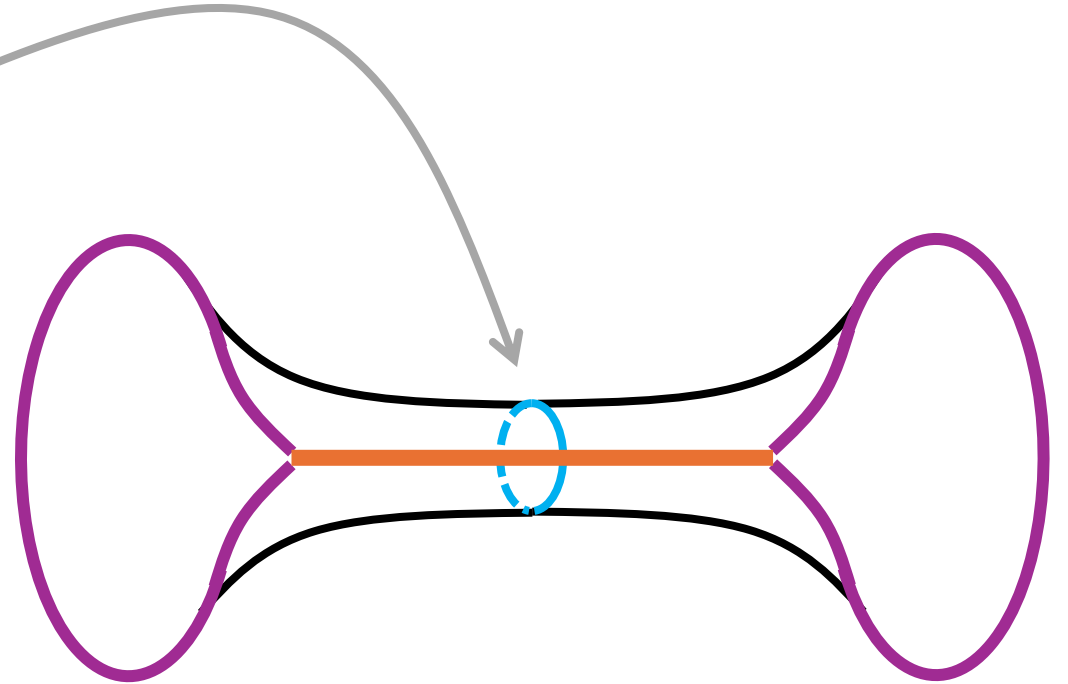
$$t \rightarrow i\tau: a(\tau)^2 = \cosh \tau/t_*$$

[Maldacena-Maoz, McInnes, Antonini-
Waddle-Simidzija-Van Raamsdonk-S, ...]

Today: a simple 2d universe and wormhole



The space of the closed universe consists of a mostly empty **circle** with a **heavy clump of matter**



Euclidean wormhole with two **asymptotic regions**

Semi-classical bulk picture

JT gravity coupled to matter, action: $I = I_{\text{JT}} + I_{\text{matter}}$

$$I_{\text{JT}} = -S_0 \chi[\Sigma] - \frac{1}{2} \int_{\Sigma} \Phi (R + 2) - \int_{\partial \Sigma} \Phi_b (K - 1)$$

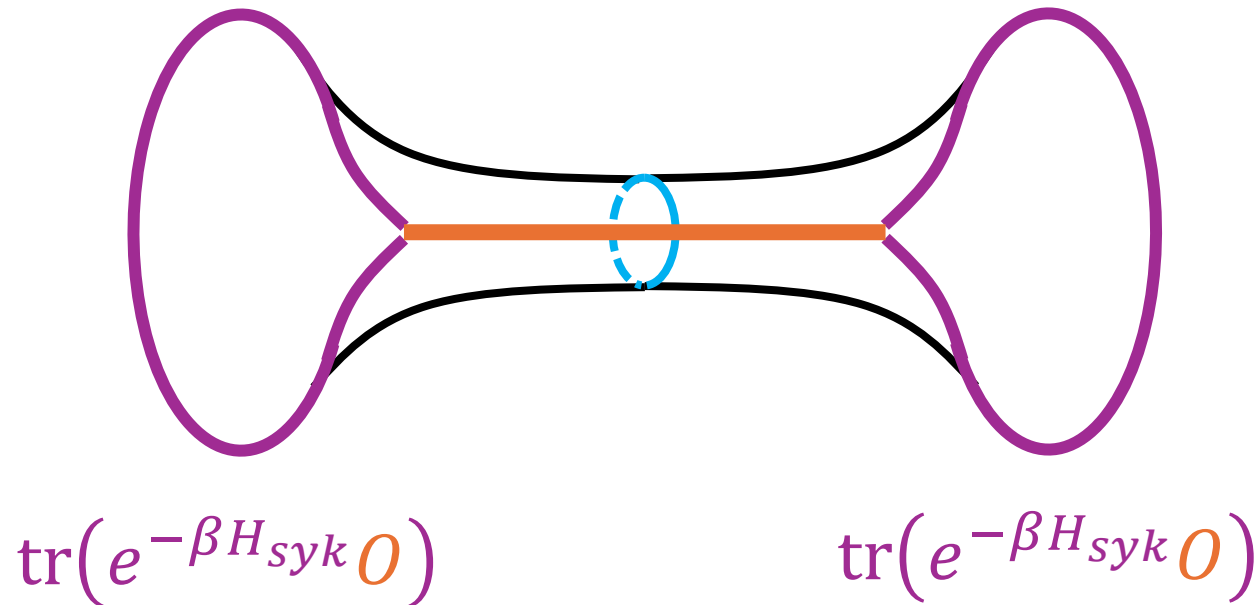
Field content:

- The metric is fixed to be constant curvature,
- The dilaton (Φ) can have a non-trivial profile
- The matter can be taken to be $O(N)$ massive fermion fields

[Kitaev, Sachdev, Maldacena-Stanford, Rosenhaus-Polchinski,...]

Microscopic proposal

- The Euclidean wormhole gives a contribution to a statistical second moment (because of the two boundaries) of some erratic data in the dual theory
- We will make this precise using the SYK model



Microscopic dual

We consider a system of N Majorana fermions χ_i

Majorana strings, a complete basis of Hermitian operators: $\mu(a) \propto \chi_1^{a_1} \cdots \chi_N^{a_N}$

Thermal 1-point functions, these completely determine the state: $\xi(a) = \frac{\text{tr} \left(e^{-\beta H} \mu(a) \right)}{\text{tr} \left(e^{-\beta H} \right)}$

We will consider ensembles of Hamiltonians and investigate the statistical properties of these 1-point functions

Sachdev-Ye-Kitaev model ($q=4$)

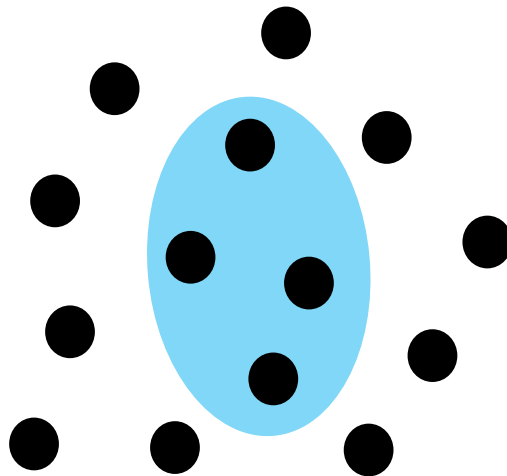
$$\chi_a^\dagger = \chi_a, \quad a = 1, \dots, N$$

$$H = \frac{1}{4!} \sum_{abcd} J_{abcd} \chi_a \chi_b \chi_c \chi_d$$

$$\{\chi_a, \chi_b\} = \delta_{a,b}$$

$$\overline{J_{abcd}} = 0, \quad \overline{J_{abcd}^2} = \frac{3! J^2}{N^3}$$

$$D = 2^{N/2}$$

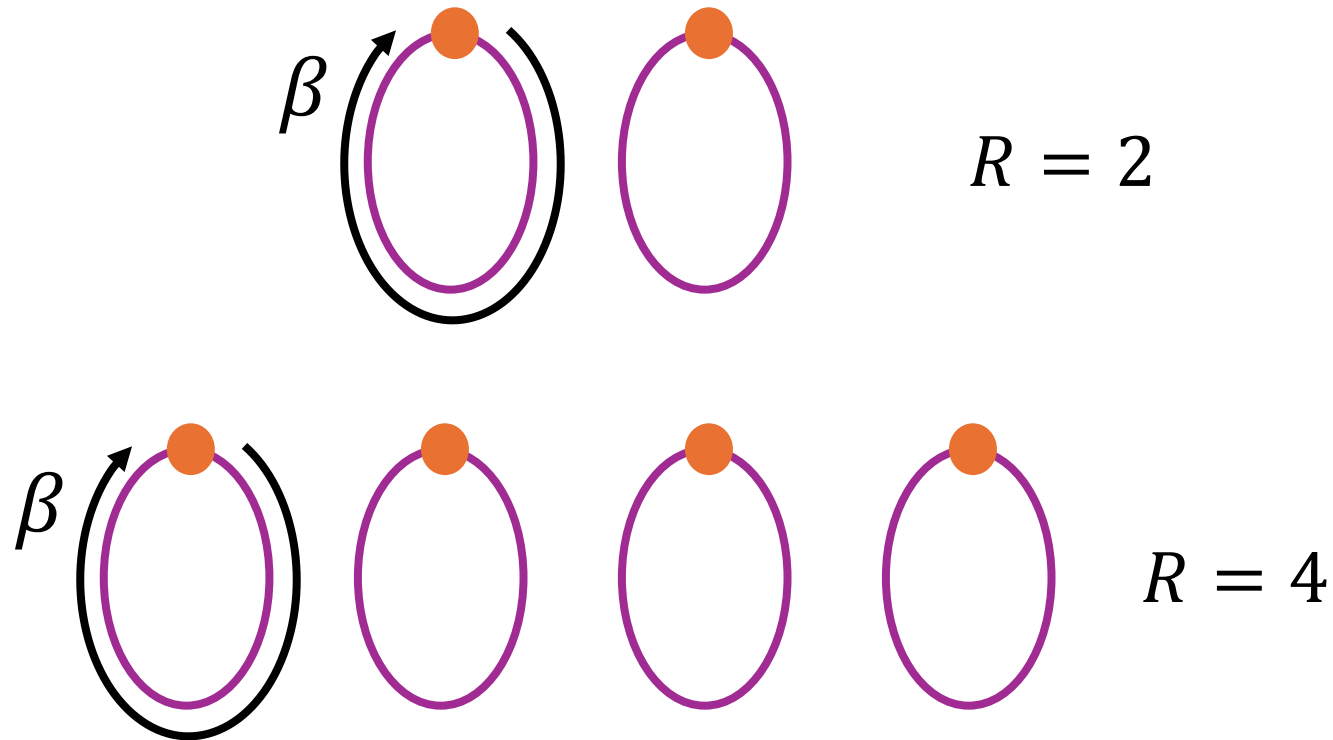


[random 2-body ensemble, Sachdev-Ye, Kitaev, Rosenhaus-Polchinski, Maldacena-Stanford, Garcia-Garcia-Verbaarschot, Bagrets-Altland-Kamenev, ...]

Statistical moments

$$\xi(a) = \frac{\text{tr} \left(e^{-\beta H} \mu(a) \right)}{\text{tr} \left(e^{-\beta H} \right)}$$

- We will compute the ensemble average of the R-th moment of the thermal 1-point function with a path integral on R replicas



Path integral computation

$$\begin{aligned}\mathrm{tr} \left(\mu(a) e^{-\beta H_{\mathrm{SYK}}} \right)^R &= \mathrm{tr} \left[\mu^{\otimes_f R} \exp(-\beta H_{\mathrm{SYK}})^{\otimes_f R} \right] \\ &= \mathrm{tr} \left[\mu^{(1)}(a) \mu^{(2)}(a) \cdots \mu^{(R)}(a) \exp \left(-\beta \sum_{r=1}^R H_{\mathrm{SYK}}^{(r)} \right) \right] \\ &\simeq \mathrm{tr} \left[(-1)^{a_1 F_{R,1}} \cdots (-1)^{a_N F_{R,N}} \exp \left(-\beta \sum_{r=1}^R H_{\mathrm{SYK}}^{(r)} \right) \right]\end{aligned}$$

$$(-1)^{F_{R,i}} := (2i)^{R/2} \chi_i^{(1)} \cdots \chi_i^{(R)}$$

Path integral computation

$$G_{rs}(\tau, \tau') := \frac{1}{N} \sum_{i=1}^N \chi_i^{(r)}(\tau) \chi_i^{(s)}(\tau'), \quad G_{rs}(\tau, \tau') = -G_{sr}(\tau', \tau),$$

multi-replica collective field

$$\overline{\text{tr}(\mu(a) e^{-\beta H_{\text{SYK}}})^R}$$

$W = \text{weight} = \# \text{ of non-identity ops in } \mu$

$$= \int [\mathcal{D}\Sigma] [\mathcal{D}G] \text{pf}[I_R \otimes \mathcal{D}^- - \Sigma]^{N-W} \text{pf}[I_R \otimes \mathcal{D}^+ - \Sigma]^W$$

$$\exp \left(-\frac{N}{2} \sum_{r,s=1}^R \int_0^\beta d\tau d\tau' \left(\Sigma_{rs}(\tau, \tau') G_{rs}(\tau, \tau') - \frac{J^2}{q} G_{rs}(\tau, \tau')^q \right) \right)$$

ensemble average washes out the dependence
on which fermions were chosen for μ

[Bettaque-S]

Path integral computation

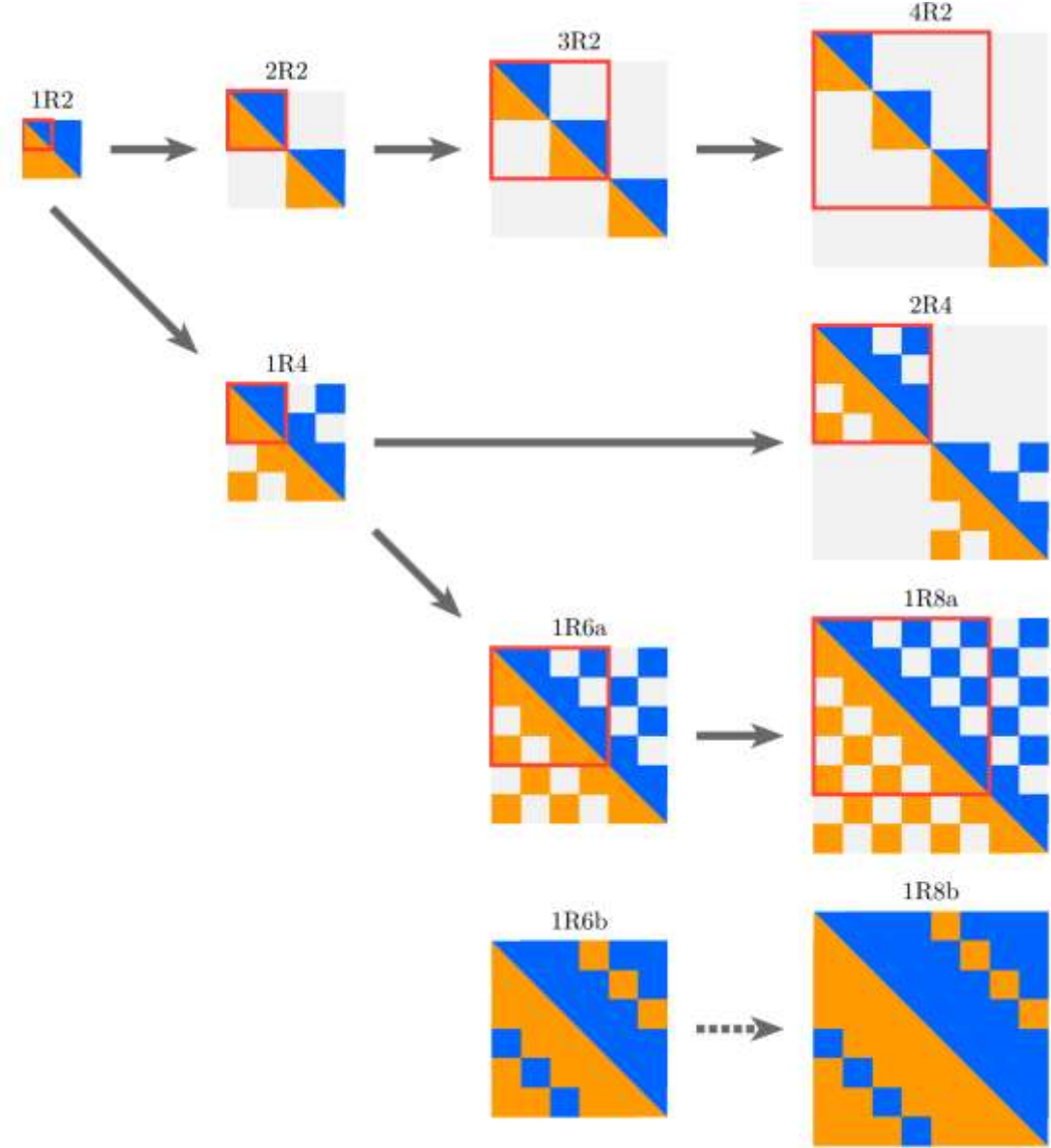
- We evaluate the path integral at large N by finding the leading saddles, but the numerics are demanding (replicas, broken time-translation, ...)

$$\mathcal{I}_{\text{SYK}}^{(R,w)}[G, \Sigma] = - (1 - w) \ln \text{pf}[I_R \otimes \mathcal{D}^- - \Sigma] - w \ln \text{pf}[I_R \otimes \mathcal{D}^+ - \Sigma] \\ + \frac{1}{2} \sum_{r,s=1}^R \int_0^\beta d\tau d\tau' \left(\Sigma_{rs}(\tau, \tau') G_{rs}(\tau, \tau') - \frac{J^2}{q} G_{rs}(\tau, \tau')^q \right)$$

$$\overline{\xi(w)^R} \approx \exp \left(-N \left[\mathcal{I}_{\text{SYK}}^{(R,w)} - R \mathcal{I}_{\text{SYK}}^{(1,0)} \right] \right)$$

$$\mathcal{D}^\mp = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & \pm 1 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{pmatrix}$$

R	Structure	G_1	G_2	G_3	G_4	G_5	G_6	G_7	Comments
2	$\mathbf{R}_2(+)$	$1/2$							
4	$\mathbf{R}_4(0,+)$	0	$1/2$	0					$\cong 2 \times \mathbf{R}_2(+)$
	$\mathbf{R}_4(+,0)$	$1/2$	0	$1/2$					
	$\mathbf{R}_4(+,+)$	$1/2$	$1/2$	$1/2$					$\rightsquigarrow \mathbf{R}_4(+,0)$ limit. convergence
6	$\mathbf{R}_6(0,0,+)$	0	0	$1/2$	0	0			$\cong 3 \times \mathbf{R}_2(+)$
	$\mathbf{R}_6(0,+,0)$	0	$1/2$	0	$1/2$	0			limit. convergence
	$\mathbf{R}_6(0,+,+)$	0	$1/2$	$1/2$	$1/2$	0			$\rightsquigarrow \mathbf{R}_6(+,+, -)$
	$\mathbf{R}_6(+,0,0)$	$1/2$	0	0	0	$1/2$			$\rightsquigarrow \mathbf{R}_6(+,0,+)$
	$\mathbf{R}_6(+,0,+)$	$1/2$	0	$1/2$	0	$1/2$			
	$\mathbf{R}_6(+,0,-)$	$1/2$	0	$-1/2$	0	$1/2$			limit. convergence
	$\mathbf{R}_6(+,+,0)$	$1/2$	$1/2$	0	$1/2$	$1/2$			$\rightsquigarrow \mathbf{R}_6(+,+, -)$
	$\mathbf{R}_6(+,+,+)$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$			$\rightsquigarrow \mathbf{R}_6(+,0,+)$
	$\mathbf{R}_6(+,+, -)$	$1/2$	$1/2$	$-1/2$	$1/2$	$1/2$			
8	$\mathbf{R}_8(0,0,0,+)$	0	0	0	$1/2$	0	0	0	$\cong 4 \times \mathbf{R}_2(+)$
	$\mathbf{R}_8(0,0,+,0)$	0	0	$1/2$	0	$1/2$	0	0	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(0,0,+,+)$	0	0	$1/2$	$1/2$	$1/2$	0	0	$\rightsquigarrow \mathbf{R}_8(+,-,+,+)$
	$\mathbf{R}_8(0,+,0,0)$	0	$1/2$	0	0	0	$1/2$	0	$\cong 2 \times \mathbf{R}_4(+,0)$
	$\mathbf{R}_8(0,+,0,+)$	0	$1/2$	0	$1/2$	0	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(0,+,0,0)$
	$\mathbf{R}_8(0,+,+,0)$	0	$1/2$	$1/2$	0	$1/2$	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(0,+,+,+)$	0	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(0,+,+,-)$	0	$1/2$	$1/2$	$-1/2$	$1/2$	$1/2$	0	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(+,0,+,0)$	$1/2$	0	$1/2$	0	$1/2$	0	$1/2$	
	$\mathbf{R}_8(+,0,+,+)$	$1/2$	0	$1/2$	$1/2$	$1/2$	0	$1/2$	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(+,+,+,0)$	$1/2$	$1/2$	$1/2$	0	$1/2$	$1/2$	$1/2$	$\rightsquigarrow \mathbf{R}_8(+,+,+, -)$
	$\mathbf{R}_8(+,+,+,+)$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	$1/2$	$\rightsquigarrow \mathbf{R}_8(+,0,+,0)$
	$\mathbf{R}_8(+,+,+,-)$	$1/2$	$1/2$	$1/2$	$-1/2$	$1/2$	$1/2$	$1/2$	



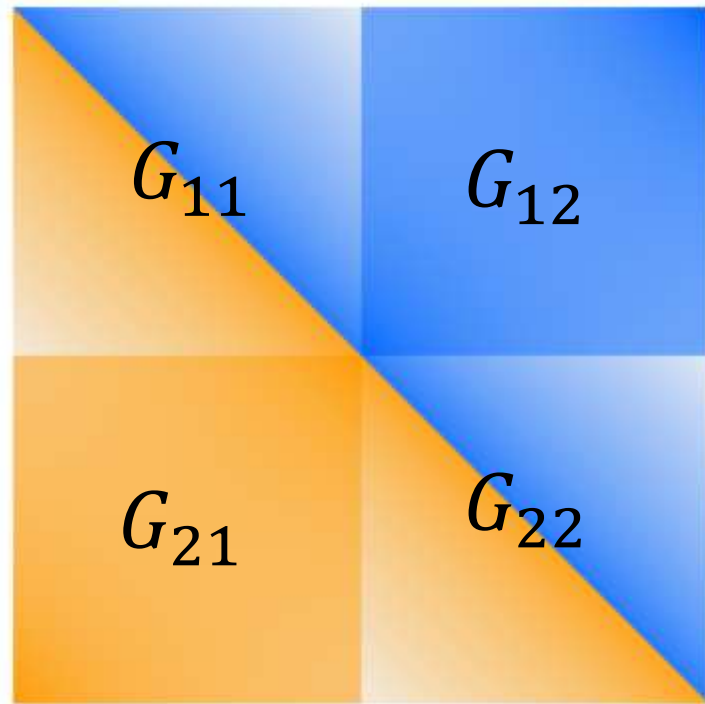
Path integral results

R=4



- We find the 1-point functions are **real gaussian** at large N provided $q > 2$ (SYK interaction degree)

R=2

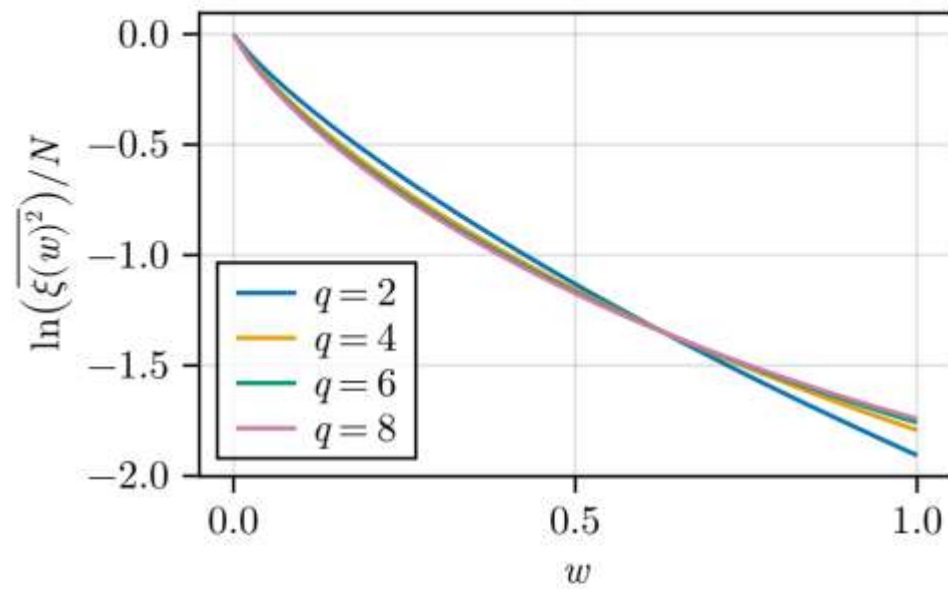


(b) $\beta = 5, w = 0.5$

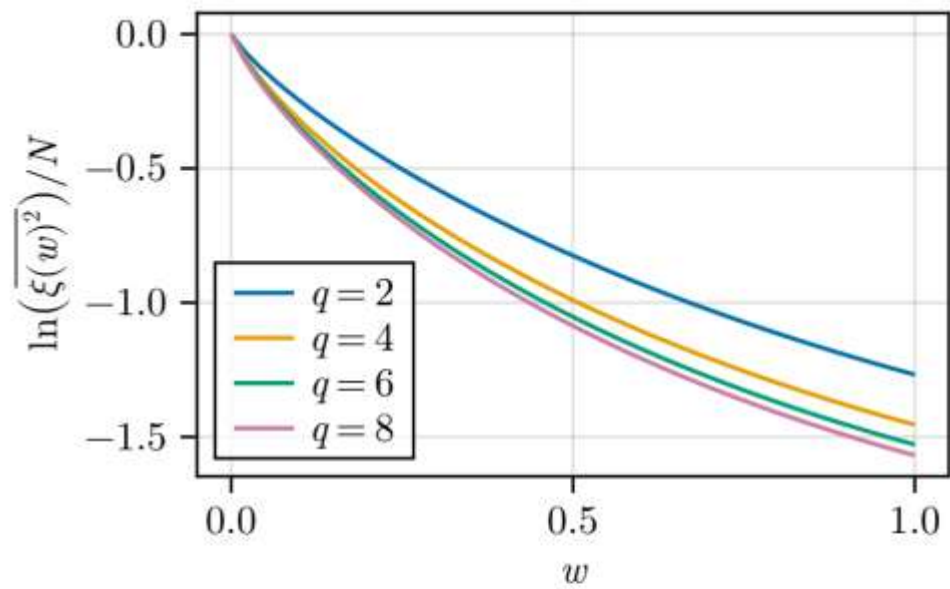


(e) $\beta = 20, w = 0.5$

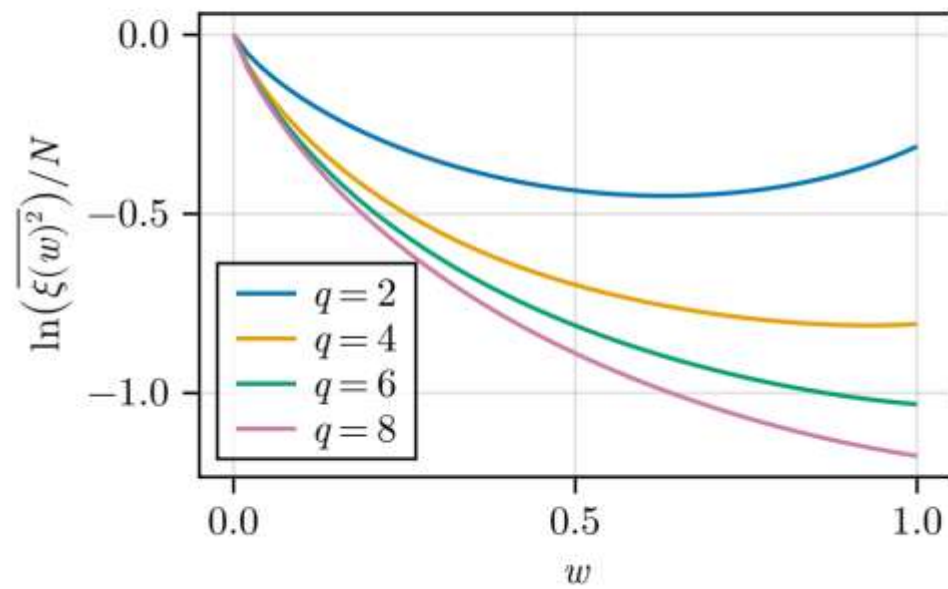
[Bettaque-S]



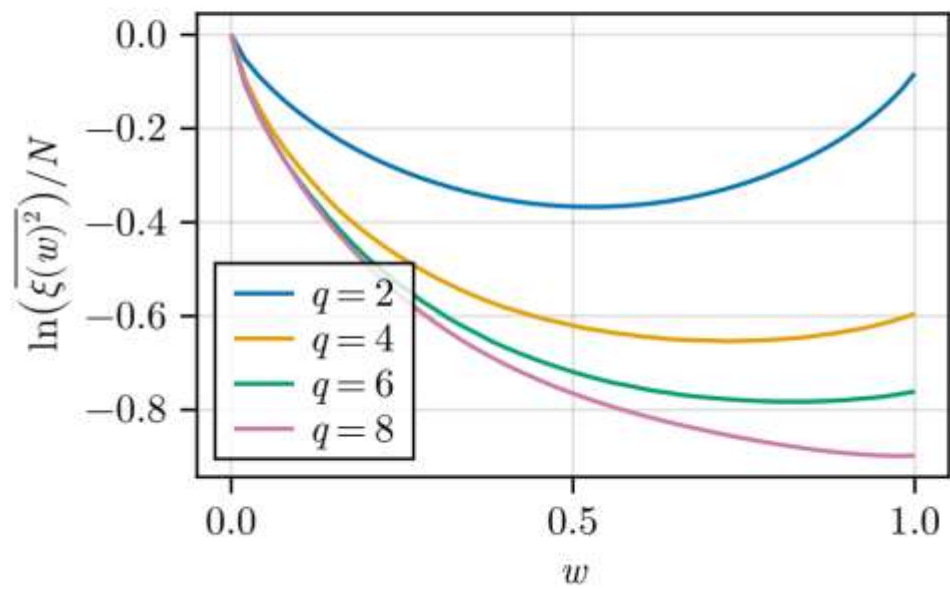
(a) $\beta = 0.5$



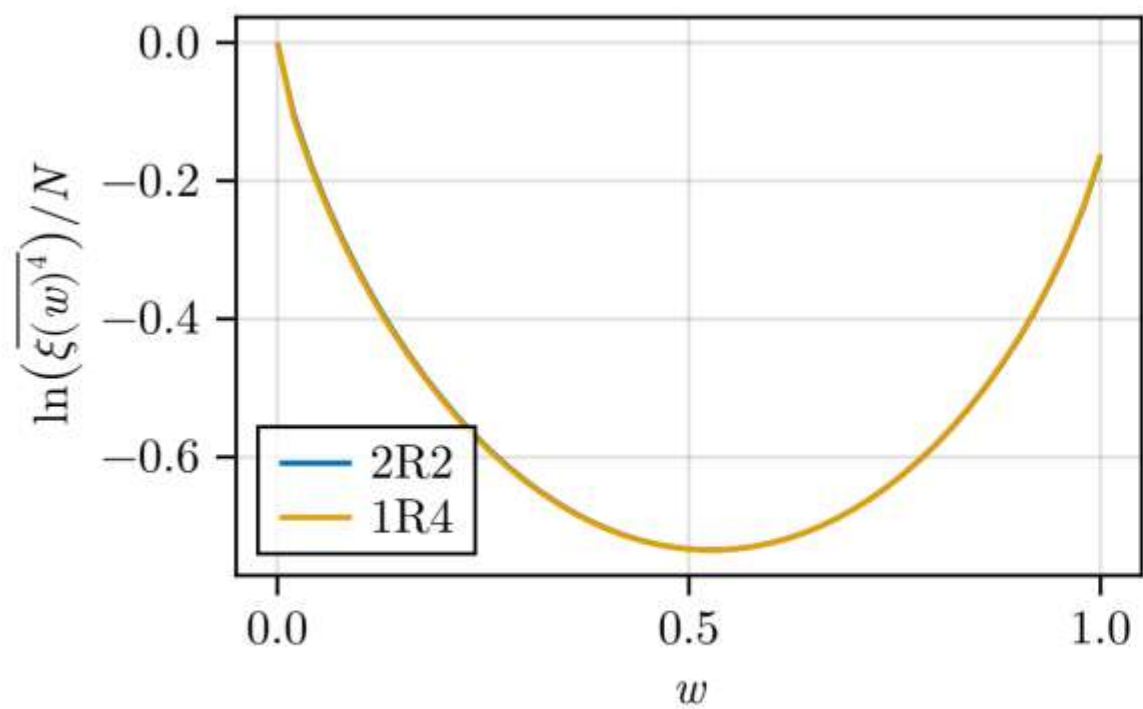
(b) $\beta = 1$



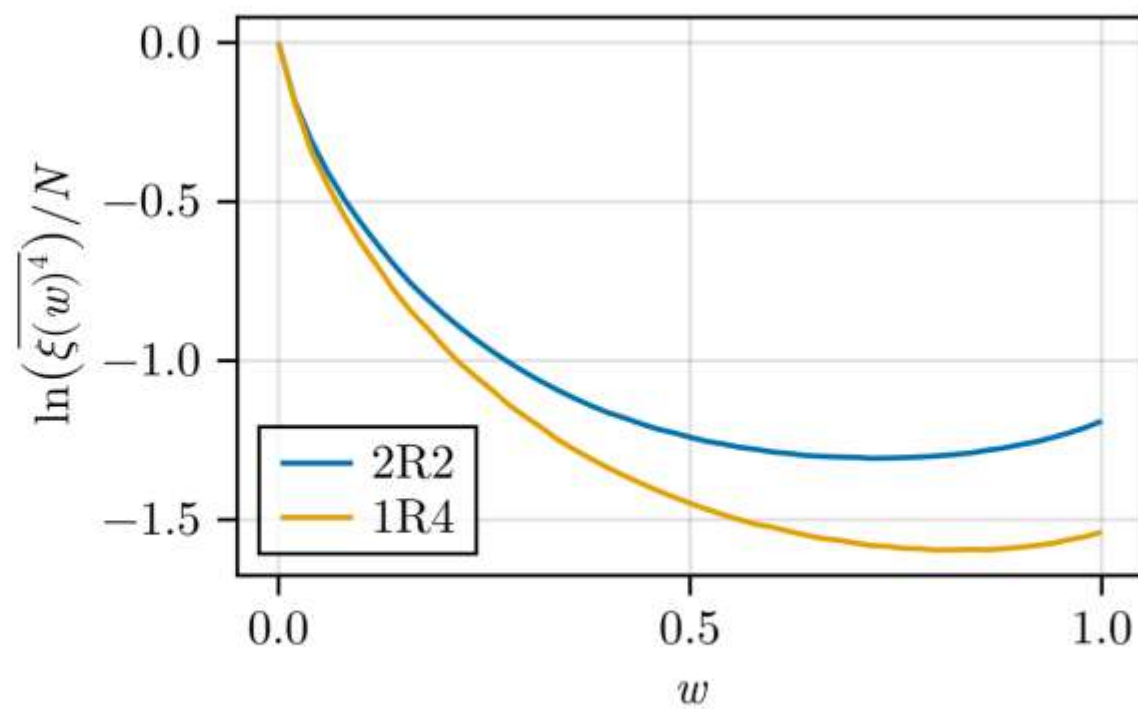
(c) $\beta = 5$



(d) $\beta = 20$



(a) $q = 2$

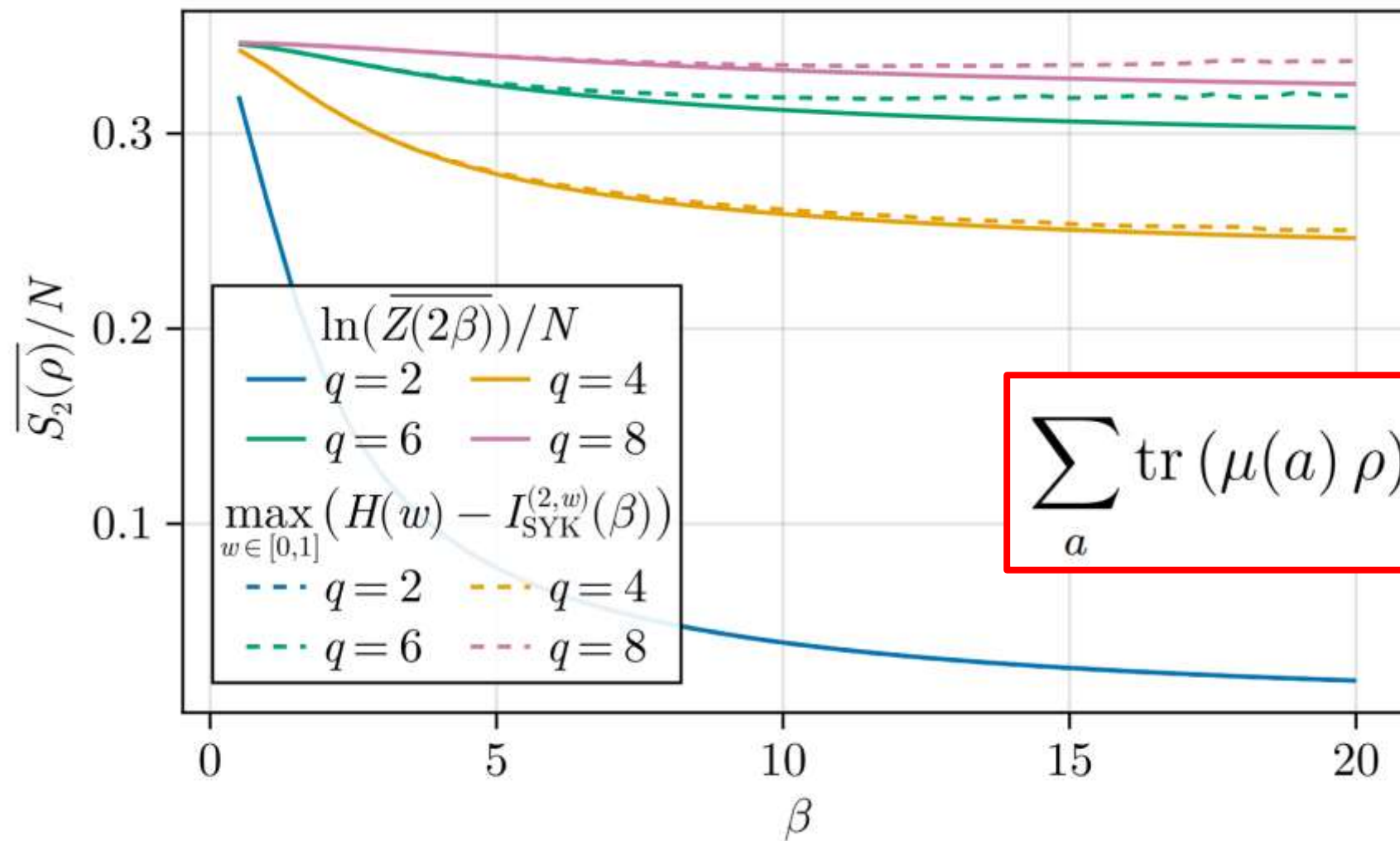


(b) $q = 4$

enhanced $O(R)$ replica symmetry

R=2 and purity

$$\overline{\sum_a \text{tr} (\mu(a) e^{-\beta H_{\text{SYK}}})^R} \approx \int_0^1 dw \exp \left(N \left[H(w) - \mathcal{I}_{\text{SYK}}^{(R,w)} \right] \right)$$



$$\sum_a \text{tr} (\mu(a) \rho)^2 = D \text{tr}(\rho^2)$$

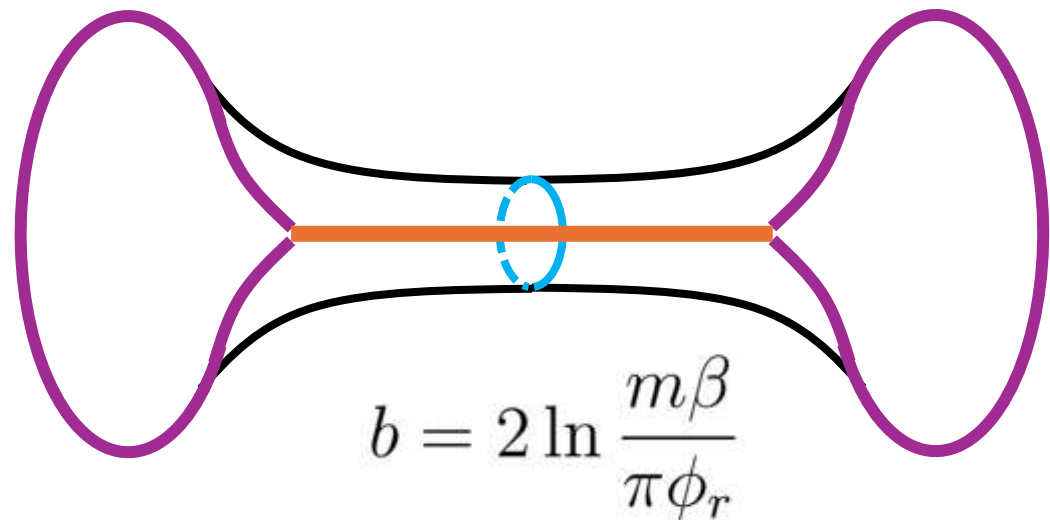
Back to the wormhole ...

- We model the SYK inter-replica correlation with a boundary-to-boundary correlations of N light probe fermions moving in a background geometry stabilized by a heavy particle [Bettaque-S]
- The bulk 2-point function of the light fermions is computed by summing over geodesics with mass $\rightarrow$ scaling dimension
- Background – double trumpet

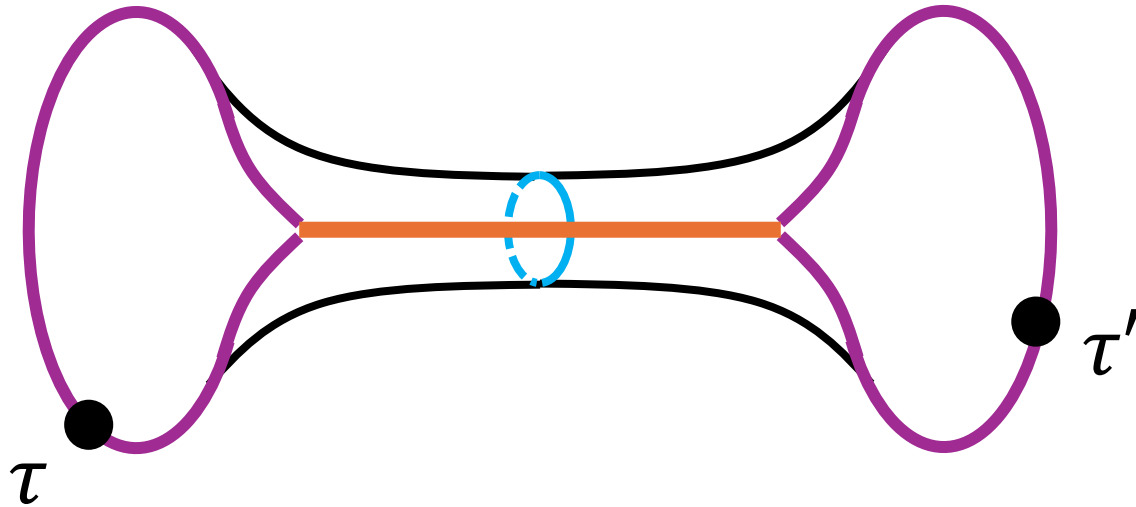
$$ds^2 = d\rho^2 + b^2 \cosh^2 \rho d\sigma^2$$

$$\Phi = \frac{\pi\phi_r}{\beta} \cosh \rho \cosh b\sigma.$$

[Stanford, Usatyuk-Wang-Zhao]



Wormhole calculation



$$e^{\rho_c} = \frac{2\beta}{\pi\epsilon} \frac{1}{\cosh b\sigma}.$$

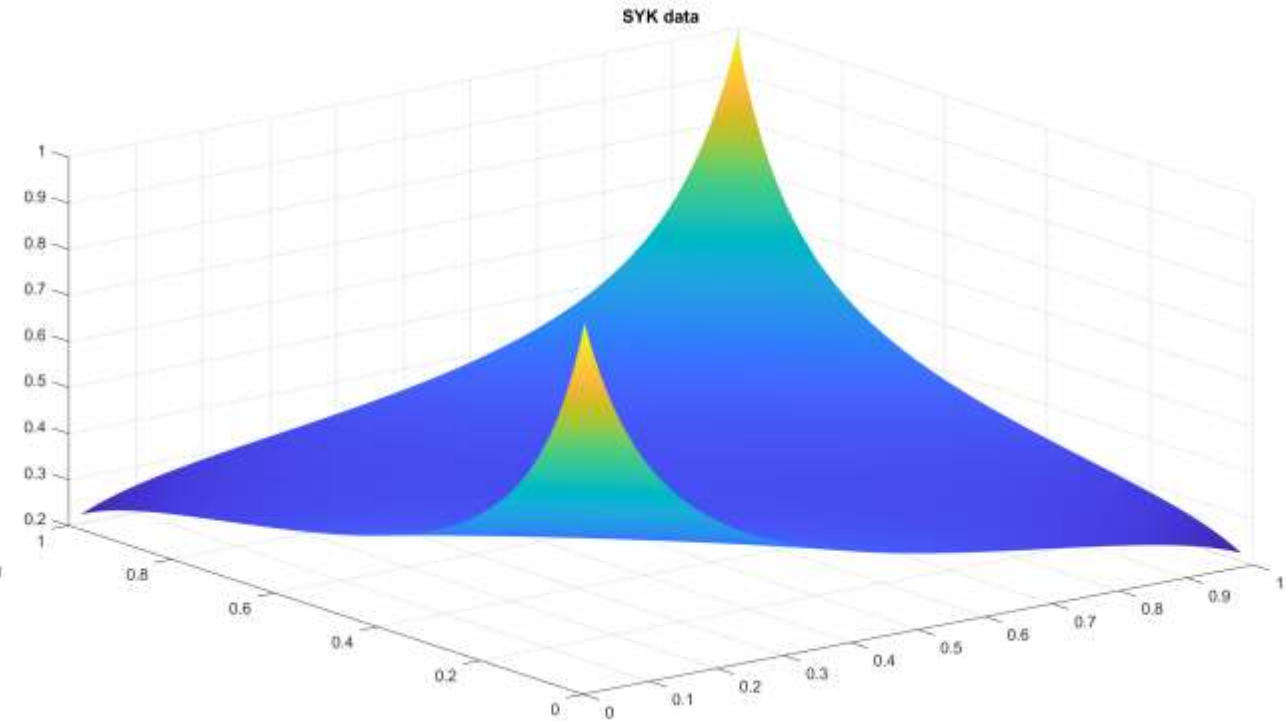
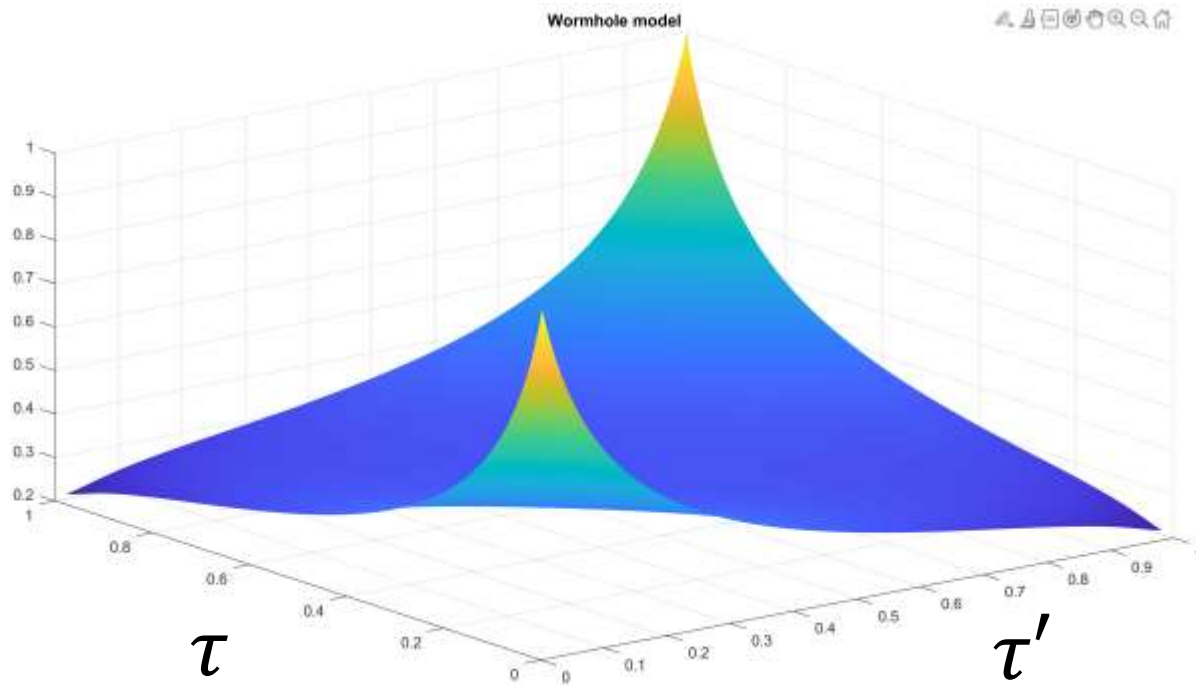
$$\frac{\tau(\sigma)}{\beta} = \frac{\arctan e^{b\sigma} - \arctan e^{-b/2}}{\arctan e^{b/2} - \arctan e^{-b/2}}.$$

winding geodesics $d_n(\sigma, \sigma') = \rho_c(\sigma) + \rho_c(\sigma') + \log \cosh^2 \frac{b(\sigma - \sigma' - n)}{2}.$

$$\mathcal{G}_P(\tau, \tau') = \mathcal{N}_P \sum_n e^{-\Delta d_n(\sigma(\tau), \sigma'(\tau'))}, \quad \mathcal{G}_{AP}(\tau, \tau') = \mathcal{N}_{AP} \sum_n (-1)^n e^{-\Delta d_n(\sigma(\tau), \sigma'(\tau'))},$$

Comparison of normalized correlations

$$\mathcal{G}_f = (1 - f)\mathcal{G}_{AP} + f\mathcal{G}_P.$$



Bulk model: $b = 6.4, f = .32$; SYK: $w = .25, \beta J = 20, L = 1000$

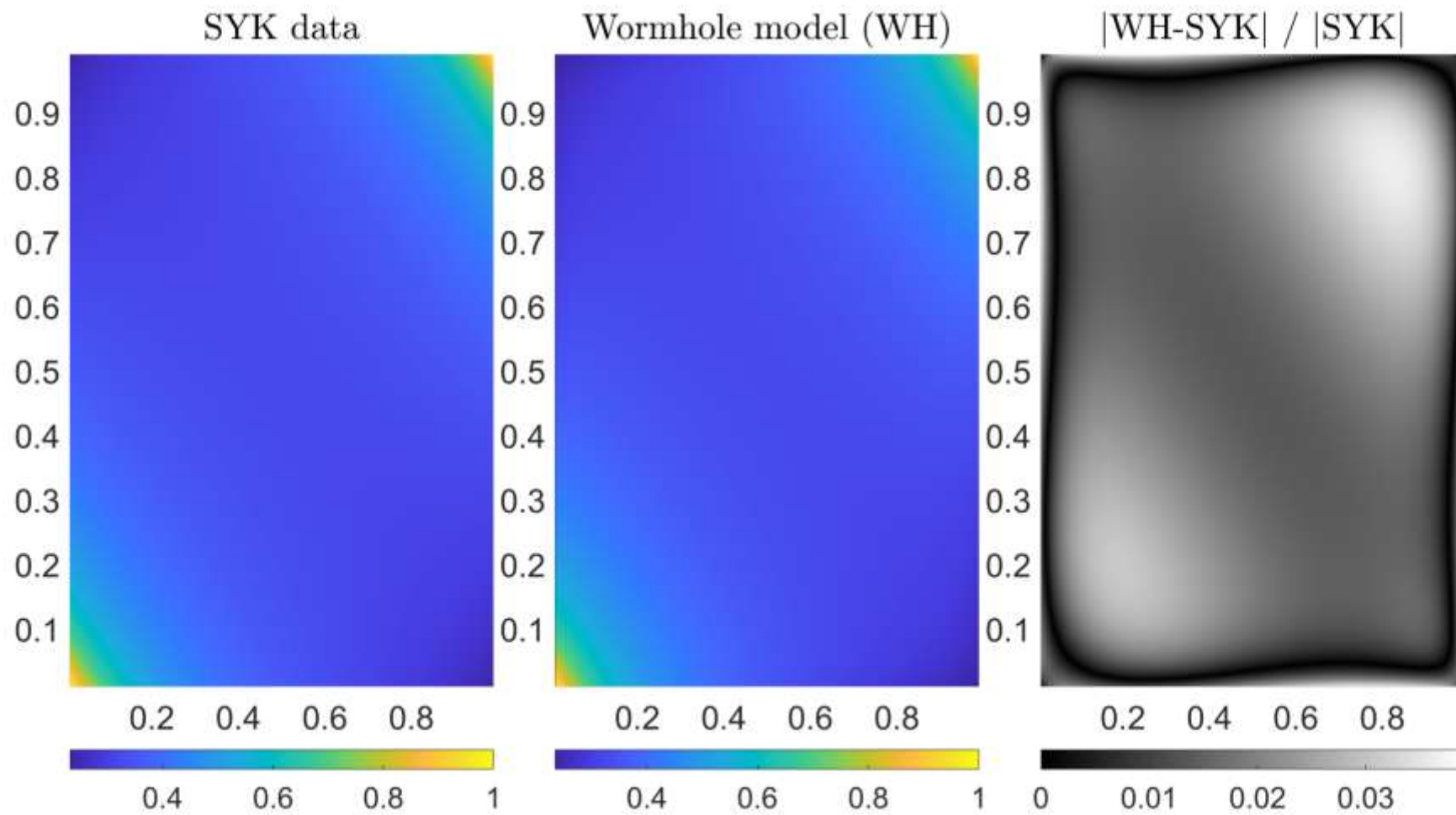
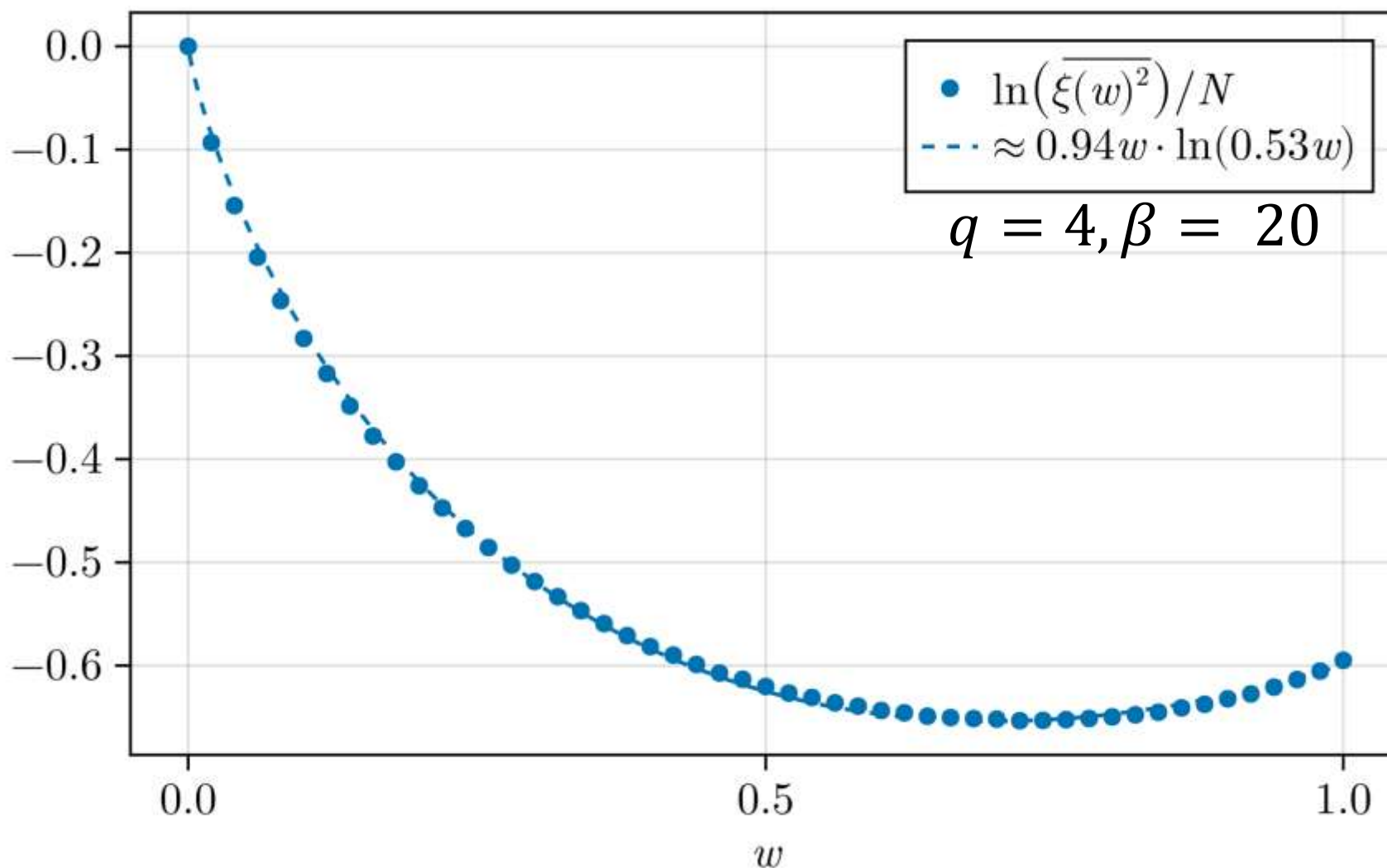


Figure 21: (Left) The plot shows the value of $G_{21}(\tau, \tau')$ in color as a function of τ/β and τ'/β . It is normalized by its maximum value, which occurs in the corners of the square. The SYK parameters are $\beta = 20$, $w = .25$, $L = 2000$. (Middle) The same configuration now showing the normalized wormhole prediction obtained by fitting to the SYK data. The fit values are $b = 6.5$ and $f = .33$. (Right) The absolute value of the relative difference between (Left) and (Middle). The maximum deviation is around 4%.

Action data

$$I_{\text{JT}} = -S_0\chi - \frac{1}{2} \int d^2x \sqrt{g} \Phi (R + 2) + (\text{boundary terms}) \textcolor{red}{- m\ell_H} + I_{\text{matter}}$$



Quantum magic

Many-body magic

- The idea of magic is relatively old, going back to [Bravyi-Kitaev](#), and is well-studied in the quantum information community
- In 2020, many-body magic was studied for the first time in the 3-state Potts model [[Cao-S-White](#)] and beyond [[Liu-Winter](#)]
- There is now an explosion of interest, including the development of novel measures such as stabilizer Renyi entropy [[Leone-Oliviero-Hamma](#), ...], magic in SYK [[Bera-Schiro](#), [Jasser-Odavic-Hamma](#), [Zhang-Zhou-Sun](#), ...], and proposed connections to gravitational backreaction in holography [[Cao-Cheng-Hamma-Leone-Munizzi-Oliviero](#)]

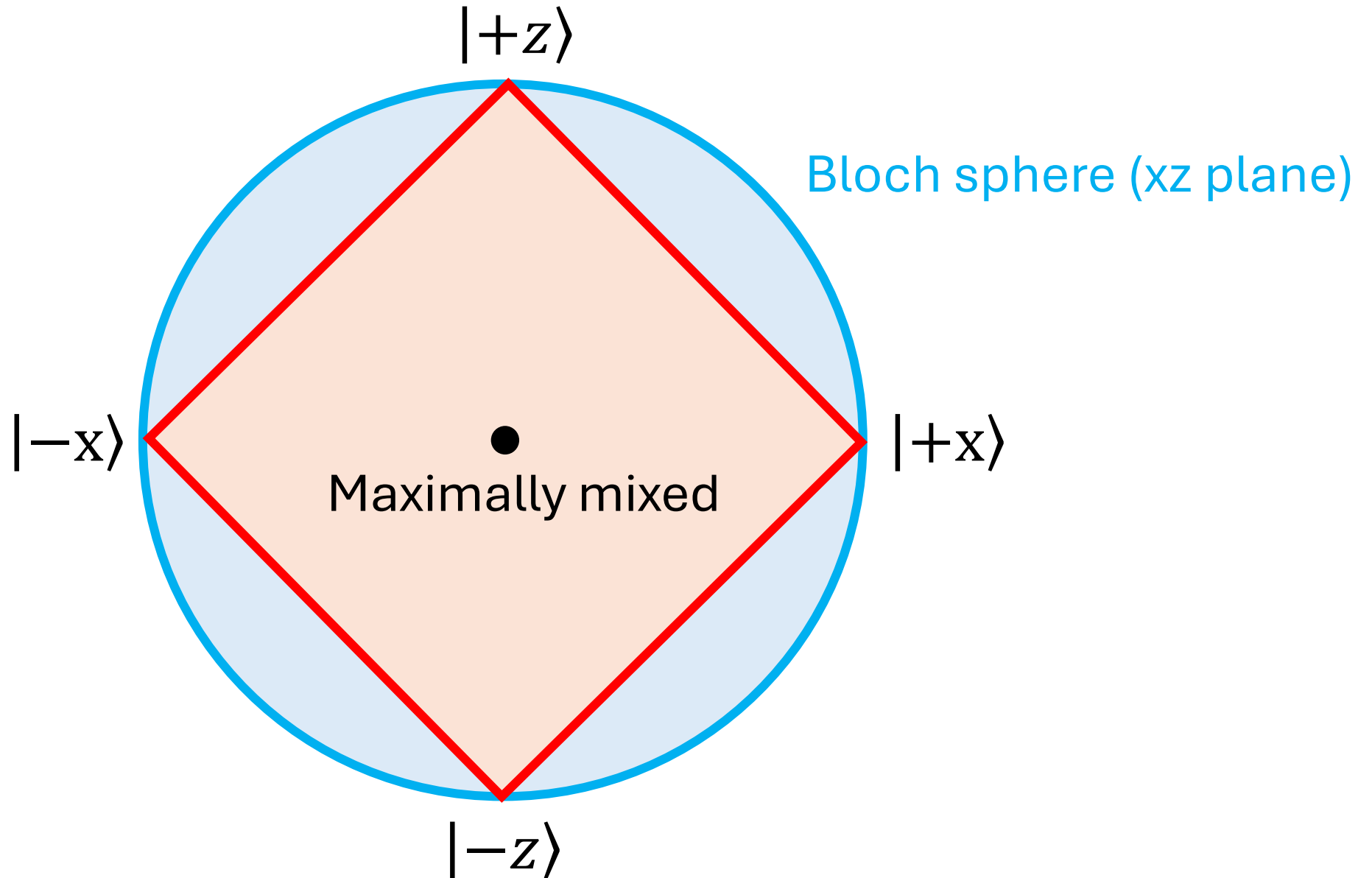
What are stabilizer states and Clifford gates?

- Clifford unitaries: the subgroup of unitaries that maps strings of Pauli operators to other strings (**no superpositions allowed**)
- Stabilizer states: mixtures of $(\text{Clifford})|0\rangle^{\otimes n}$
- Why do we care?
 - Commuting projector Hamiltonians, e.g. toric code
 - Quantum error correction
 - Random circuits and measurement induced dynamics
 - Efficient classical simulation, i.e. “low-magic” simulations

Resources in quantum theory

- Entanglement is necessary for teleportation, multi-player quantum games, and general quantum computations, and it is related to the geometry of spacetime in holography
- Low entanglement offers opportunities for classical simulation that we can take advantage of with tensor networks
- One can view entanglement as an example of a resource
 - Free states = unentangled states
 - Resource measures = entanglement entropy, ...
- Magic is the resource theory of “non-stabilizerness”

Single qubit **stabilizer polytope** (STAB = mixtures of stabilizer states):



Fermionic Cliffords and Stabilizers

Clifford group maps strings
to strings, no superposition:

$$\mu(a) \propto \chi_1^{a_1} \cdots \chi_N^{a_N}$$
$$\mu(a) \rightarrow U\mu(a)U^\dagger \propto \mu(a')$$

Stabilizer states are obtained by acting on a fixed product basis $(2i\chi_{2j-1}\chi_{2j} = 1)$ with Cliffords, which imparts classical simulability

Recently, we proved a number of facts about this group, which turns out to be $O(N, Z_2)$ after demanding fermion parity symmetry; we also showed that it forms 3-design and gave an efficient sampling algorithm

[Bettaque-S 2407.11319]

Measures of magic

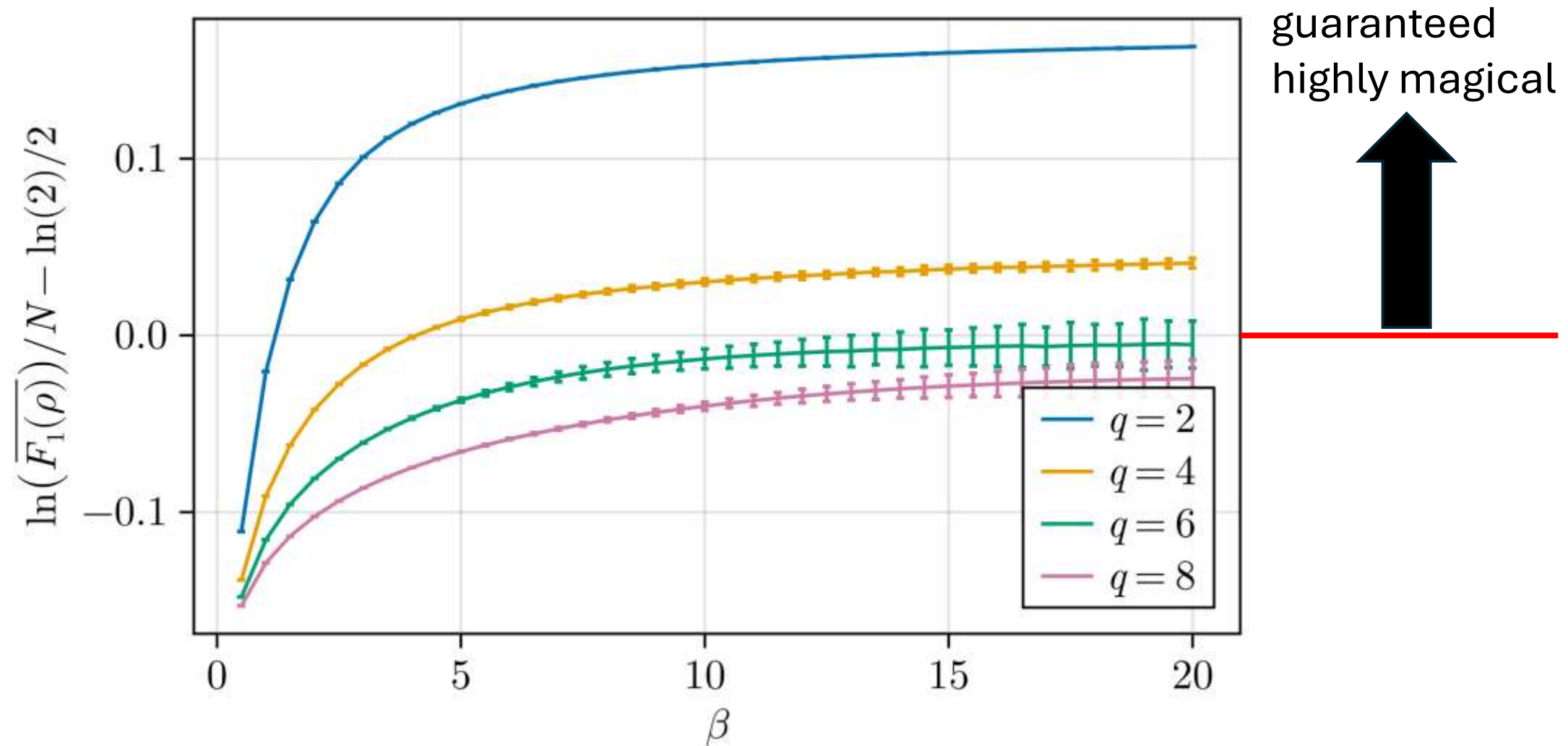
- There are many measures of magic, which tend to involve analogues of distance to the stabilizer polytope (STAB)
- Many of these are not efficiently computable, and the situation is also harder when dealing with mixed states
- We will focus for simplicity on one measure, **robustness of magic**

$$R(\rho) = \min \left(\sum_{\alpha} |c_{\alpha}| : \rho = \sum_{\alpha} c_{\alpha} \sigma_{\alpha}, \sigma_{\alpha} \in \text{STAB} \right)$$

$$R(\rho) \geq \frac{F_1(\rho)}{D}, F_1(\rho) = \sum_a |\xi(a)|$$

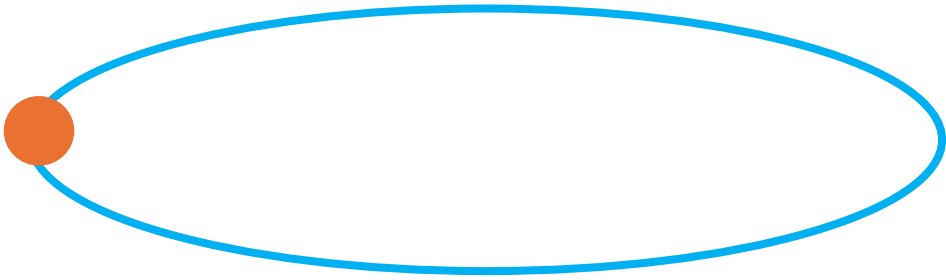
Robustness of magic

$$R(\rho) \geq \frac{F_1(\rho)}{D}, F_1(\rho) = \sum_a |\xi(a)|$$



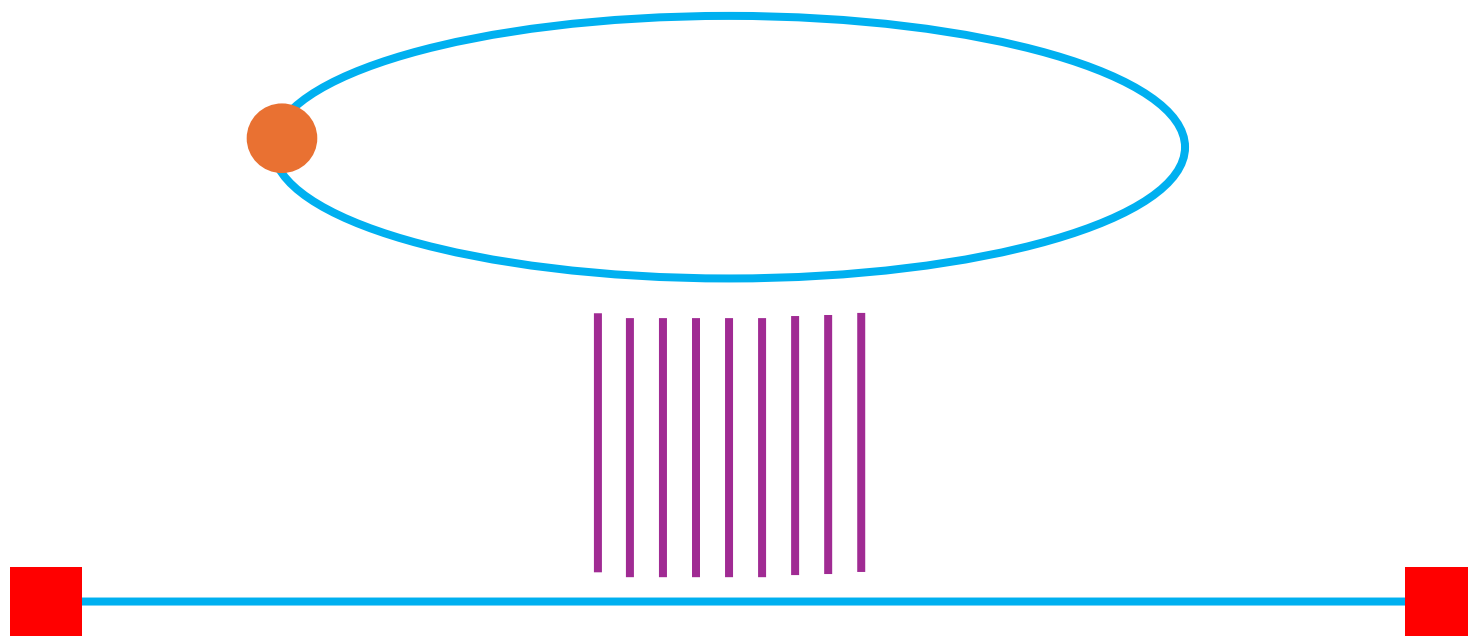
Entangling with a closed universe

The baby universe again



The space of the closed universe consists of a mostly empty **circle** with a **heavy clump of matter**





There are also quantum fields in the closed universe which can be **entangled** with fields in a separate spacetime region with **asymptotic AdS boundaries** → this gives a state in the Hilbert space associated with the boundaries

[a version of AS^2 : **Antonini-Sasieta-S**, ...]

We get two important things from this construction:

- Lots of entanglement, enabling us to connect to better studied aspects of holography like entanglement wedge reconstruction
- A richer closed universe, with the corresponding semiclassical physics now encoded in random quantum states rather than just random numbers

Microscopic model

- We start with the Sachdev-Ye-Kitaev (SYK) model on N fermions:

$$\{\psi_i, \psi_j\} = 2\delta_{ij}$$

$$H_{\text{SYK}} = i^{\frac{q}{2}} \sum_{i_1 < \dots < i_q} J_{i_1 \dots i_q} \psi_{i_1} \dots \psi_{i_q}$$



$$\overline{J_{i_1 \dots i_q} J_{i'_1 \dots i'_q}} = \frac{\mathcal{J}^2 N}{2q^2 \binom{N}{q}} \delta_{i_1, i'_1} \dots \delta_{i_q, i'_q}$$

Microscopic model

- We also need two additional ingredients:
 - (i) A first order phase transition to effectively evaporate a black hole, thus disconnecting its interior which becomes the closed universe
 - (ii) A heavy operator to support the closed universe

For (i), we use the **Maldacena-Qi** Hamiltonian:

$$H = H_{\text{SYK,L}} + H_{\text{SYK,R}}^* - \frac{\mu N}{\binom{N}{p}} (-i)^{p^2} \sum_{i_1 < \dots < i_p} \psi_{i_1 \dots i_p}^{\text{L}} \psi_{i_1 \dots i_p}^{\text{R}}$$

For (ii), we use a many-fermion operator:

$$\mathcal{O}_{\Delta} = i^{\frac{r}{2}} \psi_{i_1} \dots \psi_{i_r}$$



[Sasieta-S-Vilar Lopez]

Microscopic state

heavy operator acting on left side of
maximally entangled LR state

$$|\mathcal{O}_\Delta\rangle \equiv \mathcal{O}_\Delta^L |I\rangle$$

- Thermal pure state:

$$|\Psi_\beta^\Delta\rangle \propto e^{-\frac{\beta}{2}H} |\mathcal{O}_\Delta\rangle = \sum_n e^{-\frac{\beta}{2}E_n} c_n |E_n\rangle$$

thermal cooling with
MQ Hamiltonian

erratic coefficients

energy eigenstates
of MQ Hamiltonian

- This is the microscopic state of the coupled SYK dots; it varies erratically with the SYK couplings and the choice of operator and its statistical properties encode the closed universe physics

What do we compute?

$$\mathcal{N}_\beta^\Delta = \langle \mathcal{O}_\Delta | e^{-\beta H} | \mathcal{O}_\Delta \rangle = \sum_n e^{-\beta E_n} |c_n|^2$$

ensemble average $\overline{\mathcal{N}_\beta^\Delta} \sim e^{-I_\beta[\Sigma_d]} Z_d + e^{-I_\beta[\Sigma_h]} Z_h$

We can develop a path integral expression for this quantity either directly in the SYK language or using a corresponding bulk theory

Semi-classical bulk picture

JT gravity coupled to matter, action: $I = I_{\text{JT}} + I_{\text{matter}}$

$$I_{\text{JT}} = -S_0 \chi[\Sigma] - \frac{1}{2} \int_{\Sigma} \Phi (R + 2) - \int_{\partial \Sigma} \Phi_b (K - 1)$$

Field content:

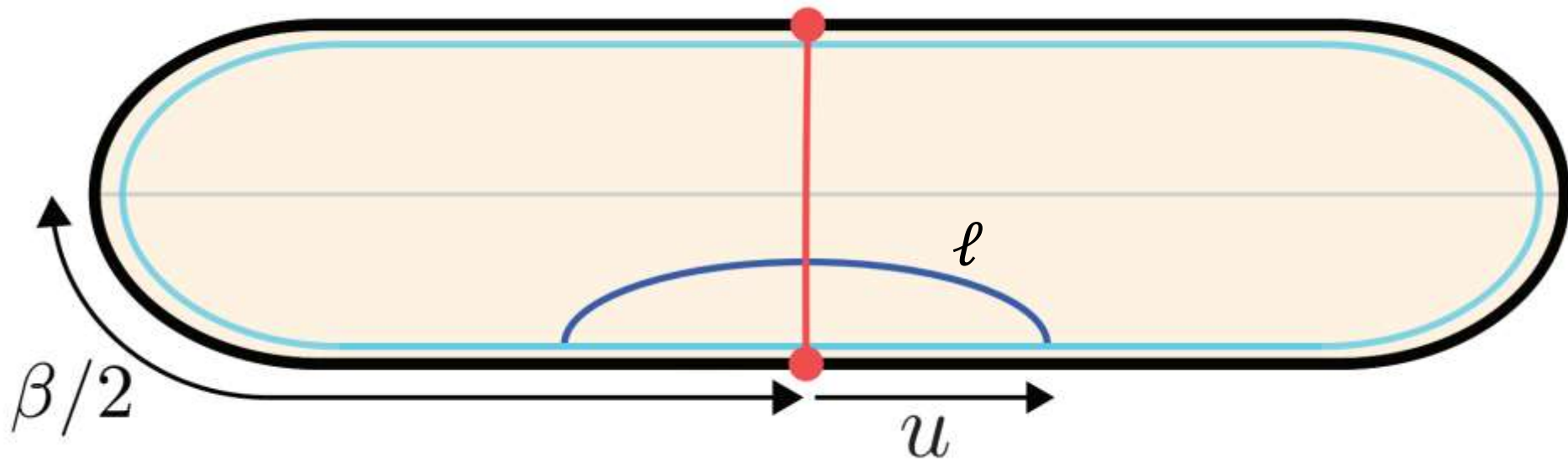
- The metric is fixed to be constant curvature,
- The dilaton (Φ) can have a non-trivial profile
- The matter can be taken to be $O(N)$ massive fermion fields

[[Kitaev](#), [Sachdev](#), [Maldacena-Stanford](#), [Rosenhaus-Polchinski](#),...]

Two saddles

$$\overline{\mathcal{N}}_{\beta}^{\Delta} \sim e^{-I_{\beta}[\Sigma_d]} Z_d + e^{-I_{\beta}[\Sigma_h]} Z_h$$

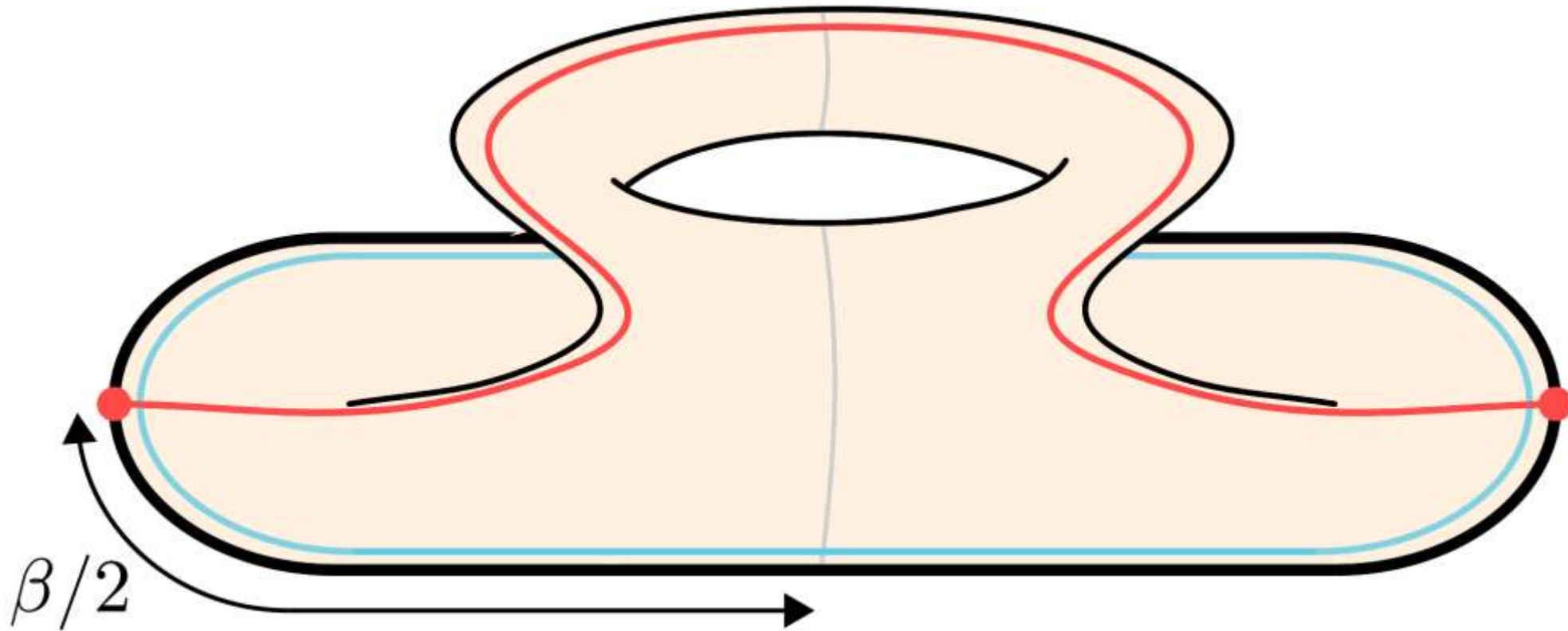
- Black hole saddle (Euclidean)

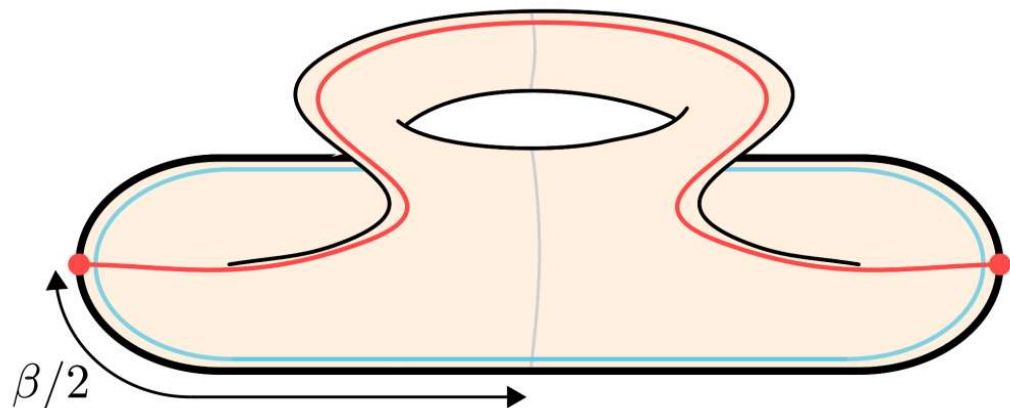


$$I = \frac{N}{2} \int du \left(\frac{1}{2} \dot{\ell}^2 + V(\ell) \right) - S_0 \quad V(\ell) = 2e^{-\ell} - \eta e^{-\Delta_H \ell} + \frac{2\Delta}{N} e^{-\ell/2}$$

[Maldacena-Qi, Sasieta-S-Vilar Lopez]

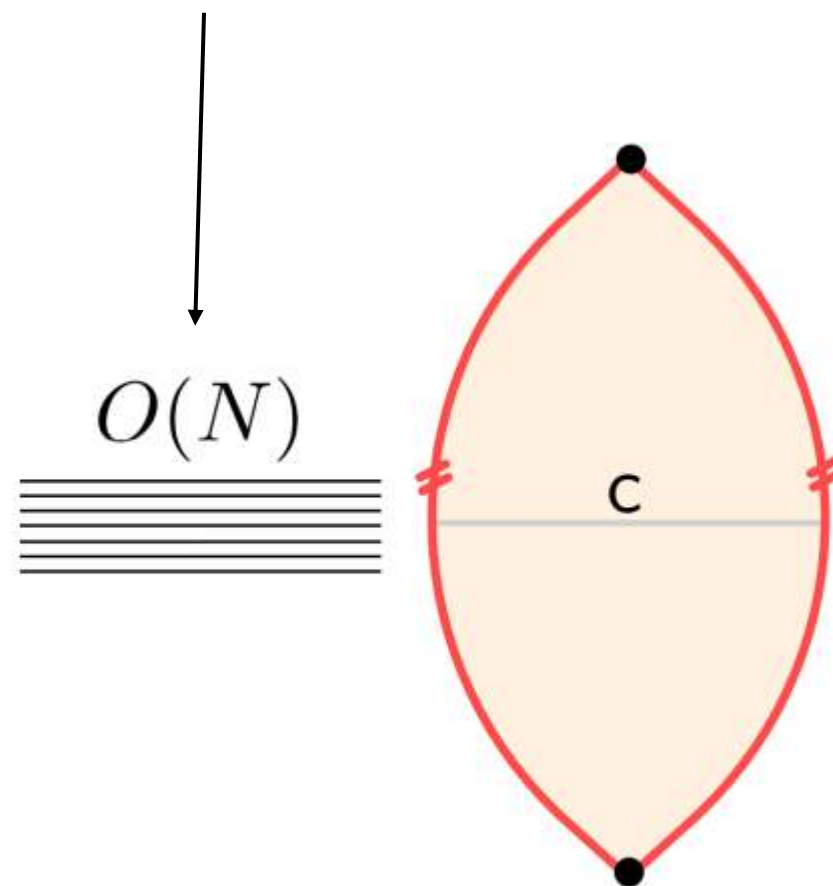
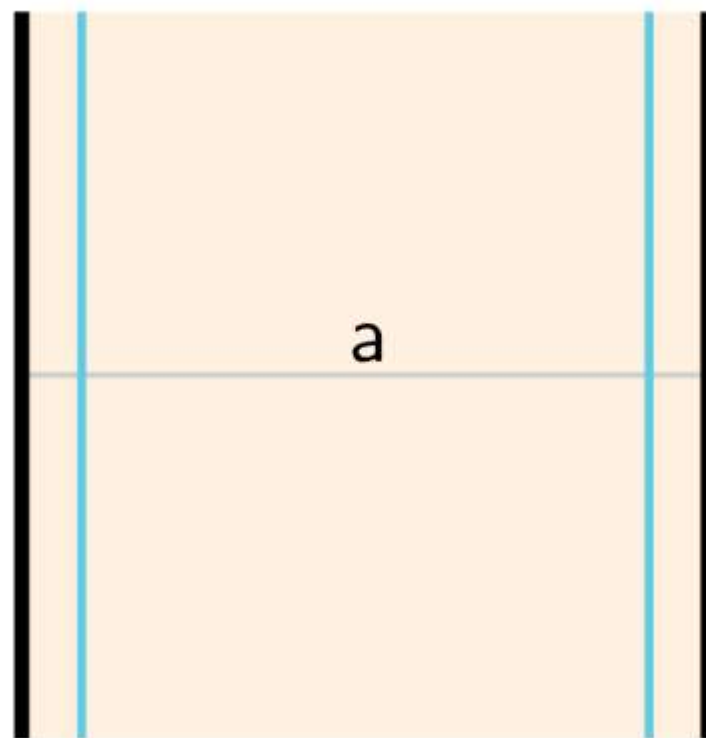
- Baby universe saddle (Euclidean)





one way to interpret the entanglement with the closed universe is in terms of the entropy of the ensemble-averaged state

Lorentzian picture:



Zero MQ temperature: an even simpler limit?

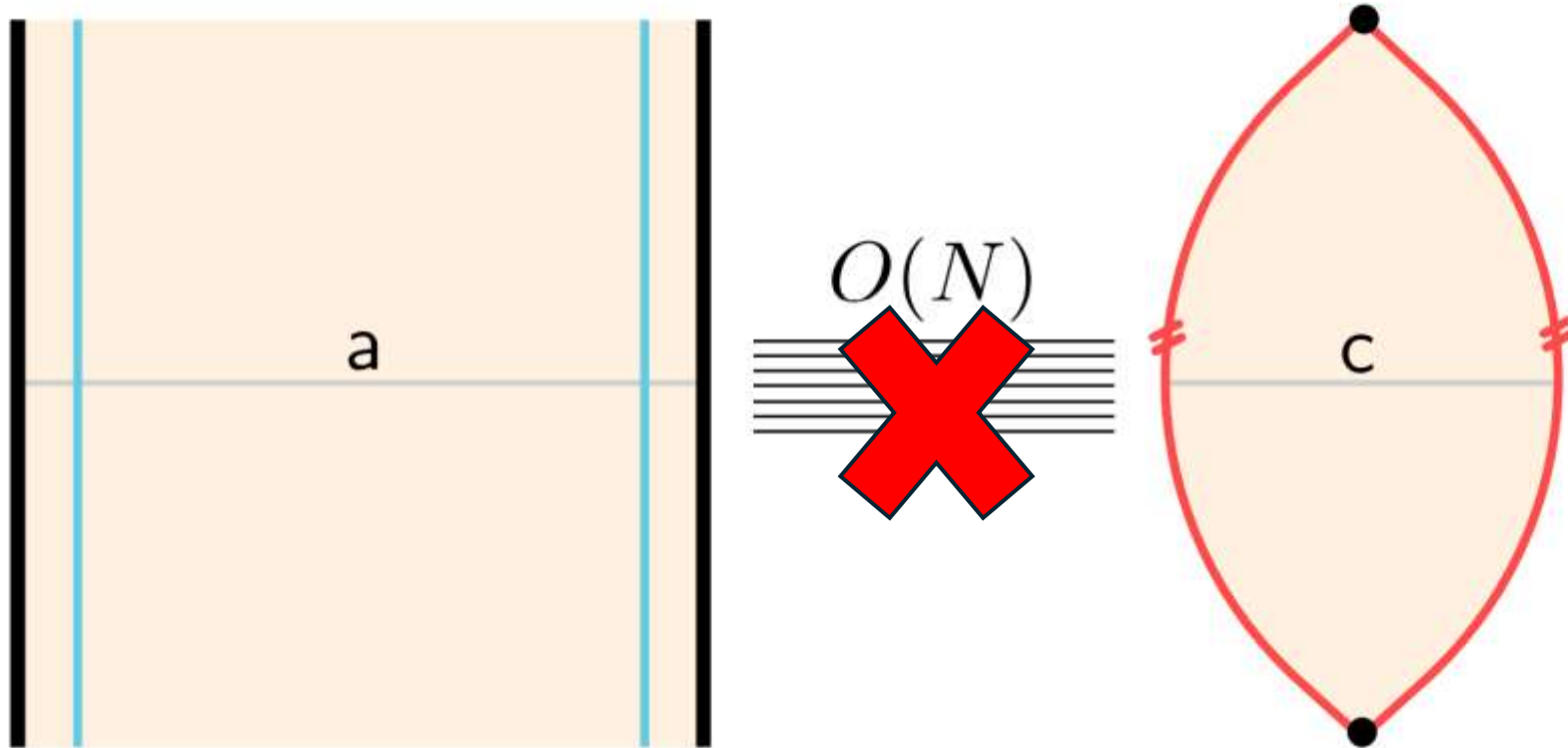
- If we take the beta parameter to infinity, then the microscopic state becomes proportional to the MQ ground state
- The MQ ground state is close to the thermofield double (TFD) state of the original SYK at a **coupling dependent temperature**

$$H = H_{\text{SYK,L}} + H_{\text{SYK,R}}^* - \frac{\mu N}{\binom{N}{p}} (-i)^{p^2} \sum_{i_1 < \dots < i_p} \psi_{i_1 \dots i_p}^{\text{L}} \psi_{i_1 \dots i_p}^{\text{R}}$$

[Maldacena-Qi, Cottrell et al.]

- The SYK TFD is not erratic and isn't dual to a closed universe???

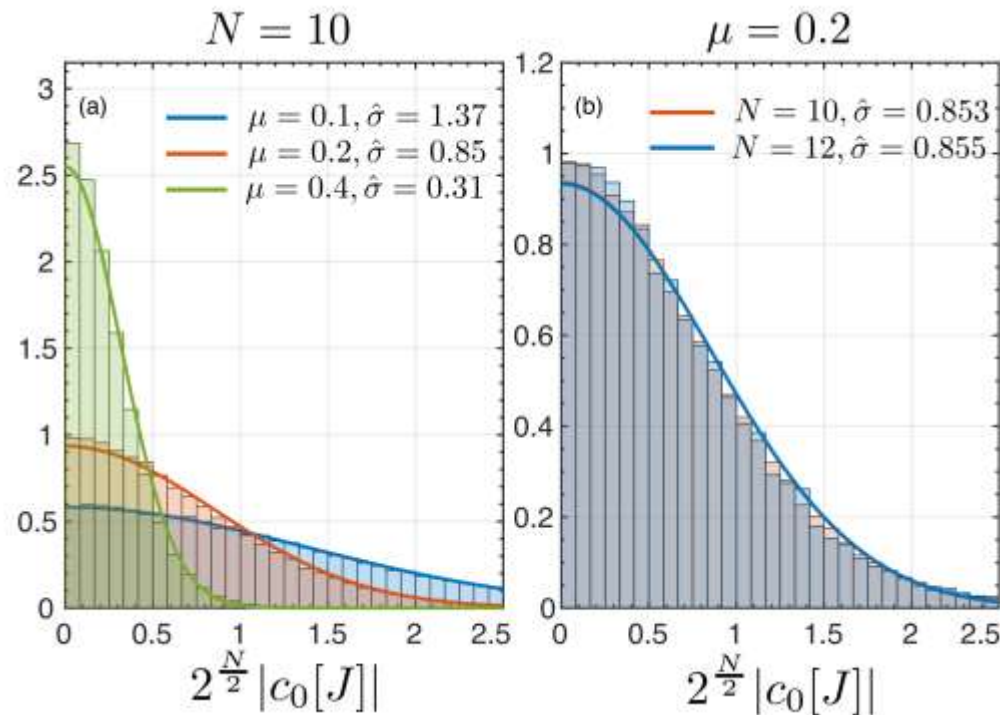
What's going on the bulk?



We're zeroing out the bulk entanglement with the closed universe, so the closed universe is floating “untethered” in the void

There is still something erratic ...

- The overlap between the MQ ground state and the heavy operator state is erratic $\rightarrow$ the closed universe is hiding in the statistical properties of this matrix element, **which is a thermal 1-pt function!**



Small N data shows that the overlap behaves as a real gaussian random variable, we saw the same at large N

[Sasieta-S-Vilar Lopez, Bettaque-S]

Outlook

Summary

- We can compute measures of magic in the SYK model using path integrals, giving one of the very few windows into magic at large N
- We found that the SYK data at low temperature could be quantitatively fit with a gravity computation involving wormholes in JT + matter, giving a kind of holographic dual of magic
- Our results also connect to holographic models of closed universes, with the SYK results providing a demonstration of the heavy operator randomness hypothesis of AS^2

Extensions and lessons (I)

It from Bit???

- We are seeing a new structure in the holographic dictionary in which averaging is essential even in higher dimensions; also manifests in other ways, e.g. a new limiting behavior in the algebraic approach
- SYK affords us the opportunity to average over Hamiltonians, but this is not essential; one can instead average over an ensemble of heavy operators in a fixed SYK realization (relevant to higher d)

[averaging/algebras: Liu, Kudler Flam-Witten, Gesteau-Engelhardt, Gesteau, observers: Engelhardt-Gesteau-Harlow, Abdalla-Antonini-Iliesiu-Levine, Harlow-Usatyuk-Zhao, saddles: Vilar Lopez-Van Raamsdonk, averaging/semiclassicality: Antonini-...-S, large q SYK ...]

Extensions and lessons (II)

- Observers are not explicitly needed (I think EFT includes this physics); one can connect to observer proposals (e.g. heavy worldline provides a reference for gravitational dressing)
- We stayed away from cosmological data in this talk, but there are directions, e.g. naturalness vs big bang, dark energy from evolving scalars, primordial black holes encoding the heavy particles associated with the erratic behavior???
- Novel connections to quantum information notions like magic, and a new holographic dual for magic

A dream:

Use a quantum computer to simulate and study toy models of quantum cosmology and the Big Bang in a holographic model of quantum gravity

A start:

$$G_{21} \sim \overline{tr[exp(-\beta H_{syk})O\chi(\tau)]tr[exp(-\beta H_{syk})O\chi(\tau')]}$$